Real Book Exercises, Lectures 1–4: Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 24 exercises of `Listas/lista-reais-lectures-01-04.pdf`

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These are the solutions to the list of 24 real book exercises for Lectures 1–4, in the list's order and with its numbering (one block of six per lecture: two foundations, three exam level, one advanced). Each exercise is cited by book, number and printed page; its statement is reproduced verbatim in the list and is not repeated here. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992) and Mas-Colell, Whinston & Green (1995) (MWG). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of iterated elimination of strictly dominated strategies (IESDS), with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $S_i^\infty = \bigcap_k S_i^k$. Mixed strategies are $\sigma_i \in \Sigma_i = \Delta(S_i)$ (Lecture 2). In bimatrices player 1 picks rows and the first payoff is player 1's. Where the book gives an answer (Muñoz-Garcia's worked answers, the MWG solutions manual) each item records whether it agrees. Every matrix, dominance claim, equilibrium and formula is checked by the Python scripts in this folder (`check_A1.py`, `check_A2.py` for items 1–12; `check_B*.py` for items 13–24).

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Lecture 1 — Static games I: dominance and iterated elimination

1 Tadelis Exercise 4.5

Foundations · block L1

Source: Tadelis Exercise 4.5, page 77 (statement in the book and in Exercise 1 of the list; not reproduced here).

What is asked. Find the strategy profiles of the given 3×3 game that survive IESDS.

Method. IESDS round by round (Lecture 1, slides 28–29; Tadelis §4.2.1, p. 65): in each round remove every strategy that is strictly dominated given the opponent's surviving strategies.

Solution. Matrix as printed in the book (p. 77):

$1 \setminus 2$	L	C	R
U	6, 8	2, 6	8, 2
M	8, 2	4, 4	9, 5
D	8, 10	4, 6	6, 7

Round 1 (S^0 = full game).

- Player 1: U is strictly dominated by M: $6 < 8$, $2 < 4$, $8 < 9$. D is not dominated by M (ties $8 = 8$ and $4 = 4$), and M, D are not dominated by U.
- Player 2: her payoffs are $L = (8, 2, 10)$, $C = (6, 4, 6)$, $R = (2, 5, 7)$ in rows (U, M, D). C is beaten by L in rows U, D but not in M; it beats R in row U. No column is strictly dominated.

$S_1^1 = \{M, D\}$, $S_2^1 = \{L, C, R\}$.

Round 2. Player 2 now compares columns on rows (M, D) only: $L = (2, 10)$, $C = (4, 6)$, $R = (5, 7)$. C is strictly dominated by R ($4 < 5$, $6 < 7$); L and R are not ranked. Player 1 on all three columns: $M = (8, 4, 9)$, $D = (8, 4, 6)$: M is better only against R, so D is weakly but not strictly dominated. $S_1^2 = \{M, D\}$, $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $M = (8, 9)$, $D = (8, 6)$, a tie against L, so no strict dominance. Player 2 on (M, D): $L = (2, 10)$, $R = (5, 7)$, not ranked. The process stops.

$$S^\infty = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Interpretation. Rationality of player 1 removes U. Then player 2's knowledge that player 1 is rational removes C, which can be a best reply only to beliefs that put positive weight on U. The iterated-elimination equilibrium is a set of four profiles, not a point. Two of them, (M, R) with payoffs $(9, 5)$ and (D, L) with $(8, 10)$, are profiles where each strategy is a best reply to the other (the pure Nash equilibria of Lecture 2). Deleting the weakly dominated D as well would single out (M, R), but that uses weak dominance, a stronger and less safe assumption than IESDS.

Verification. `check_A1.py` runs IESDS with the lecture's definition and finds exactly the rounds above (round 1: U; round 2: C; then stop), that U is dominated only by M and C only by R, and that D is weakly dominated by M in the final set.

Answer. Round 1: delete U (dominated by M). Round 2: delete C (dominated by R once U is gone). Nothing else is strictly dominated: $S^\infty = \{M, D\} \times \{L, R\}$, four profiles.

2 Gibbons Problem 1.2

Foundations · block L1

Source: Gibbons Problem 1.2, page 48 (statement in the book and in Exercise 2 of the list; not reproduced here).

What is asked. Which strategies of the given 3×3 game survive IESDS, and what are the pure-strategy Nash equilibria?

Method. IESDS (Gibbons §1.1.B, pp. 4–7; Lecture 1, slide 29). For the second question, a pure Nash equilibrium is a profile in which each player's strategy is a best reply to the other's (Gibbons §1.1.C; Lecture 2). Every Nash equilibrium survives IESDS (Gibbons Prop. B, p. 12), so we only need to search the surviving cells.

Solution. The matrix as printed (the top-left cell reads "2.0" in the scan; it is $(2, 0)$):

$1 \setminus 2$	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Round 1. Player 1: B is strictly dominated by T ($1 < 2$, $0 < 1$, $3 < 4$); it is not dominated by M ($3 > 2$ against R). T and M are best replies to R and L, so they stay. Player 2, columns over rows (T, M, B): $L = (0, 4, 3)$, $C = (1, 2, 2)$, $R = (2, 3, 0)$. C is *not* dominated yet: it beats R in row B ($2 > 0$) and L in row T. $S_1^1 = \{T, M\}$, $S_2^1 = \{L, C, R\}$.

Round 2. On rows (T, M): $C = (1, 2)$ and $R = (2, 3)$, so C is strictly dominated by R. $L = (0, 4)$ and $R = (2, 3)$ are not ranked. $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $T = (2, 4)$, $M = (3, 2)$, not ranked. Player 2 on (T, M): $L = (0, 4)$, $R = (2, 3)$, not ranked. Stop.

$$S^\infty = \{T, M\} \times \{L, R\}.$$

Pure Nash equilibria. Check the four surviving cells:

- (T, L): player 1 would switch to M ($3 > 2$). Not NE.
- (T, R): player 1 gets $4 \geq 3, 2$ in column R; player 2 gets $2 \geq 0, 1$ in row T. NE.
- (M, L): player 1 gets $3 \geq 2, 1$; player 2 gets $4 \geq 2, 3$. NE.
- (M, R): player 1 would switch to T ($4 > 2$). Not NE.

The pure Nash equilibria are (T, R) and (M, L).

Interpretation. Note the order: C is a sensible reply only to beliefs that put weight on B; once player 2 knows player 1 is rational, C goes. IESDS leaves four profiles; Nash equilibrium, which requires beliefs to be correct, cuts them to two. This previews Lecture 2.

Verification. `check_A1.py` reproduces the rounds (round 1: B; round 2: C), checks that C is not dominated in the full game, and finds by brute force over all nine cells that the pure Nash equilibria are exactly (T, R) and (M, L). Kumar's unofficial Gibbons manual (Answer 1.2, not in `Livros/`) gives the same deletions and equilibria; it writes the survivors as "(T,M) and (L,R)", meaning $\{T, M\}$ for player 1 and $\{L, R\}$ for player 2.

Answer. IESDS: delete B (by T), then C (by R); survivors $\{T, M\} \times \{L, R\}$. Pure Nash equilibria: (T, R) with payoffs $(4, 2)$ and (M, L) with payoffs $(3, 4)$.

3 Tadelis Exercise 4.6

Exam · block L1

Source: Tadelis Exercise 4.6, page 77 (statement in the book and in Exercise 3 of the list; not reproduced here).

What is asked. Two roommates choose cleaning times $t_i \geq 0$ with payoff $u_i = (10 - t_j)\, t_i - t_i^2$. Find (a) the best-response correspondence, (b) the choices that survive one round of IESDS, and (c) those that survive IESDS.

Method. IESDS with continuous strategies by bounding best replies, exactly as in Tadelis's Cournot example (§4.2.2, pp. 65–67, Fig. 4.1). Tool: for a payoff that is a concave parabola in own action, an action outside the range of possible best replies is strictly dominated by the nearest end of that range.

Solution. **(a) Best response.** u_i is strictly concave in t_i ($\partial^2 u_i / \partial t_i^2 = -2$). The first-order condition $10 - t_j - 2t_i = 0$ and the constraint $t_i \geq 0$ give the unique best reply

$$BR_i(t_j) = \max\left\{0,\ \frac{10 - t_j}{2}\right\} = \max\left\{0,\ 5 - \frac{t_j}{2}\right\}.$$

It is a function (single-valued) and decreasing in t_j: cleaning is a strategic substitute.

Dominance lemma. Write $x = (10 - t_j)/2$, so $u_i(t, t_j) = t(2x - t)$. For any two times a, t,

$$u_i(a, t_j) - u_i(t, t_j) = (t - a)\,(t + a - 2x). \tag{1}$$

If $t > a \geq x$ the right side is positive; if $t < a \leq x$ it is positive too. So if every possible x (given the opponent's surviving set) lies in $[\underline{a}, \bar{a}]$, then every $t > \bar{a}$ is strictly dominated by $\bar{a}$ and every $t < \underline{a}$ by $\underline{a}$. Conversely, a t that is the unique best reply to some surviving t_j is not strictly dominated.

(b) One round. With $t_j \in [0, \infty)$, the best reply ranges over $[0, 5]$ ($BR = 5$ at $t_j = 0$, $BR = 0$ for $t_j \geq 10$). By (1) with $\bar{a} = 5$, every $t_i > 5$ is strictly dominated by $t_i = 5$. Every $t_i \in [0, 5]$ is the best reply to $t_j = 10 - 2t_i \in [0, 10]$, so it survives. Hence $S_i^1 = [0, 5]$.

(c) Iterating. If $S_j^k = [L_k, U_k]$, the surviving best replies form $[BR(U_k), BR(L_k)]$, so

$$L_{k+1} = 5 - \tfrac{U_k}{2}, \qquad U_{k+1} = 5 - \tfrac{L_k}{2}, \qquad (L_1, U_1) = (0, 5).$$

k	1	2	3	4	5	∞
S_i^k	$[0, 5]$	$[2.5, 5]$	$[2.5, 3.75]$	$[3.125, 3.75]$	$[3.125, 3.4375]$	$\{10/3\}$

The width halves each round: $U_{k+1} - L_{k+1} = \frac{1}{2}(U_k - L_k)$, so $U_k - L_k = 5/2^{k-1} \to 0$. The midpoint M_k satisfies $M_{k+1} = 5 - M_k/2$, whose fixed point is $10/3$, and

$$M_k - \tfrac{10}{3} = (-\tfrac{1}{2})^{k-1}(M_1 - \tfrac{10}{3}) = (-\tfrac{1}{2})^{k-1}(-\tfrac{5}{6}) \to 0.$$

Therefore $S_i^\infty = \bigcap_k S_i^k = \{10/3\}$ for both roommates: IESDS predicts the unique profile $t_1 = t_2 = 10/3$, the point where $t = 5 - t/2$.

Interpretation. Each round uses one more layer of knowledge: round 1, "I am rational" (never clean more than 5 hours); round 2, "I know you are rational" (you clean at most 5, so I clean at least 2.5); and so on. Each player gets $u_i = (10 - 10/3)(10/3) - (10/3)^2 = 100/9$. Both cleaning $t = 2.5$ would give $(7.5)(2.5) - 6.25 = 12.5 > 100/9$: the IESDS outcome over-cleans relative to the joint optimum. Each roommate ignores that his cleaning lowers the value of the other's.

Verification. `check_A1.py` checks the best reply and the factorization (1) with sympy, runs the interval recursion for 40 rounds in exact arithmetic (width $5/2^{k-1}$ and the midpoint formula at every round), and, as a second route, runs brute-force IESDS on the grid $\{0, \frac{1}{6}, \ldots, 10\}$, which leaves exactly $t_1 = t_2 = 10/3$.

Answer. (a) $BR_i(t_j) = \max\{0, (10 - t_j)/2\}$. (b) $S_i^1 = [0, 5]$: any $t_i > 5$ is strictly dominated by 5. (c) Rounds give $[2.5, 5]$, $[2.5, 3.75]$, $[3.125, 3.75]$, …, with widths $5/2^{k-1}$; only $t_1 = t_2 = 10/3$ survives IESDS.

4 Tadelis Exercise 4.7

Exam · block L1

Source: Tadelis Exercise 4.7, page 77–78 (statement in the book and in Exercise 4 of the list; not reproduced here).

What is asked. Two candidates choose a positive (P), balanced (B) or negative (N) campaign; payoffs are win probabilities given by the book's rules. (a) Model it as a normal-form game; (b) write the matrix; (c) describe each stage of IESDS and say whether it gives a clear prediction.

Method. Normal form (slides 13–14); IESDS (slide 29; Tadelis §4.2).

Solution. **(a)** $I = \{1, 2\}$, $S_1 = S_2 = \{P, B, N\}$. Let $\pi(s_i, s_j)$ be the probability that the candidate playing s_i wins against s_j. The book's rules give

$$\pi(s, s) = 0.5, \quad \pi(P, B) = 0, \quad \pi(P, N) = 0.3, \quad \pi(N, B) = 0.6.$$

The other three entries follow because exactly one candidate wins: $\pi(B, P) = 1$, $\pi(N, P) = 0.7$, $\pi(B, N) = 0.4$. Payoffs are $u_1(s_1, s_2) = \pi(s_1, s_2)$ and $u_2(s_1, s_2) = 1 - u_1(s_1, s_2) = \pi(s_2, s_1)$. Player 2's payoff function is the same rule applied with the roles swapped. It is the complement of player 1's because the probabilities of the two candidates winning add up to one: the game is constant-sum and symmetric.

(b)

$1 \setminus 2$	P	B	N
P	0.5, 0.5	0, 1	0.3, 0.7
B	1, 0	0.5, 0.5	0.4, 0.6
N	0.7, 0.3	0.6, 0.4	0.5, 0.5

(c) IESDS.

- Round 1. Player 1's rows over columns (P, B, N): $P = (0.5, 0, 0.3)$, $B = (1, 0.5, 0.4)$, $N = (0.7, 0.6, 0.5)$. P is strictly dominated by B ($0.5 < 1$, $0 < 0.5$, $0.3 < 0.4$) and also by N. B and N are not ranked (B is better against P, N against B and N). By symmetry the same holds for player 2. $S_i^1 = \{B, N\}$.
- Round 2. On $\{B, N\}$: $B = (0.5, 0.4)$, $N = (0.6, 0.5)$. B is strictly dominated by N. The same holds for player 2. $S_i^2 = \{N\}$.

$S^\infty = \{(N, N)\}$: both run negative campaigns and each wins with probability 0.5. The prediction is clear (a unique profile) after two rounds.

Interpretation. B survives round 1 only because it crushes a positive campaign. Once each candidate knows the other will not run a purely positive campaign, going fully negative wins. The outcome has the same win probabilities as (P, P), and in a constant-sum game every profile is Pareto optimal *for the candidates*. If voters dislike negative campaigning, the equilibrium is a prisoner's-dilemma outcome for them: the candidates end up exactly where they would have been with positive campaigns, but with a nastier campaign.

Verification. `check_A1.py` builds the matrix from the four rules plus complementarity, checks symmetry, that P is dominated by both B and N, and that IESDS removes P (round 1) and B (round 2) for both players, leaving (N, N).

Answer. (a) $S_i = \{P, B, N\}$, $u_1 = \pi(s_1, s_2)$ from the rules, $u_2 = 1 - u_1$ (the same rule from candidate 2's side). (b) Matrix above. (c) Round 1 deletes P (dominated by B and N); round 2 deletes B (dominated by N); unique prediction (N, N), each wins with probability 0.5.

5 Muñoz-Garcia Ch. 1, Exercise 12

Exam · block L1

Source: Muñoz-Garcia Ch. 1, Exercise 12, page 23 (statement in the book and in Exercise 5 of the list; not reproduced here).

What is asked. For the 3×3 game of Fig. 1.30: (a) strictly dominant strategies; (b) strictly dominated strategies; (c) what each player deletes under rationality alone; (d) what survives under common knowledge of rationality (IESDS).

Method. Dominance (slide 23) and the epistemic reading of IESDS (slide 28; Tadelis §4.2.1, p. 65). Round 1 of IESDS uses only each player's own rationality. Round 2 uses "each knows the other is rational", and so on.

Solution. Matrix from the book (p. 23):

1 \ 2	F	C	B
F	0, 5	2, 4	2, 3
C	2, 3	0, 5	3, 2
B	5, 0	2, 4	2, 3

(a) Player 1's best rows are B against F (5), F or B against C (tie at 2), and C against B (3). No row is the unique best everywhere: no strictly dominant strategy. Player 2's best columns are F in row F (5), C in row C (5), C in row B (4): none is best everywhere. Neither player has a strictly dominant strategy.

(b) Player 1: $F = (0, 2, 2)$, $C = (2, 0, 3)$, $B = (5, 2, 2)$ over columns (F, C, B). B and C are the unique best replies to columns F and B, so they are not dominated; F ties with B against columns C and B and beats C against column C, so it is not strictly dominated either (it is weakly dominated by B). No row is strictly dominated. Player 2: $F = (5, 3, 0)$, $C = (4, 5, 4)$, $B = (3, 2, 3)$ over rows (F, C, B). B is strictly dominated by C: $3 < 4$, $2 < 5$, $3 < 4$. F and C are not dominated (each is the best reply to some row). The only strictly dominated strategy is player 2's B.

(c) Rationality, no common knowledge. Each player deletes his own strictly dominated strategies but does not use the fact that the other does the same. Player 2 drops B; player 1 has nothing to drop and still evaluates his rows against all three columns:

$$\{F, C, B\} \times \{F, C\} = \{(F, F), (F, C), (C, F), (C, C), (B, F), (B, C)\}.$$

(d) Common knowledge of rationality. Player 1 knows player 2 is rational, so he compares rows on columns $\{F, C\}$ only: $F = (0, 2)$, $C = (2, 0)$, $B = (5, 2)$. Now C is strictly dominated by B ($2 < 5$, $0 < 2$); F is only weakly dominated by B (tie against C). Round 3: player 2 compares $F = (5, 0)$ and $C = (4, 4)$ on rows (F, B); neither dominates. Player 1 compares $F = (0, 2)$ and $B = (5, 2)$; only weak. Stop:

$$S^\infty = \{F, B\} \times \{F, C\} = \{(F, F), (F, C), (B, F), (B, C)\}.$$

Interpretation. Rationality alone removes only strategies that are bad against *everything*. Player 1's C is a good reply to B, so it can only be discarded by someone who knows that player 2 will not play B, that is, by someone who uses player 2's rationality. Each extra layer of "I know that you know" can remove more strategies; here two layers suffice.

Verification. `check_A1.py` checks the dominance statements in (a)–(b), that IESDS removes B for player 2 in round 1 and C for player 1 in round 2, and that F is only weakly dominated by B in the final game. The book's answer (pp. 23–26) agrees with (a)–(d). Its text for (b) and (d) describes the comparisons loosely (for example, it says player 1 deletes C because his payoff "is higher from B ... when player 2 plays F"), but the deletions and the final sets are the same as ours.

Answer. (a) None. (b) Only player 2's B (strictly dominated by C). (c) Rationality alone: $\{F, C, B\} \times \{F, C\}$, six profiles. (d) With common knowledge, player 1 then drops C (dominated by B on $\{F, C\}$): $S^\infty = \{F, B\} \times \{F, C\}$, four profiles.

6 MWG Exercise 8.B.5

Advanced · block L1

Source: MWG Exercise 8.B.5, page 262 (statement in the book and in Exercise 6 of the list; not reproduced here).

What is asked. Linear Cournot with price $a - b(q_1 + q_2)$ and unit cost c: (a) show that iterated deletion of strictly dominated strategies gives a unique prediction; (b) say whether this remains true with three firms.

Method. IESDS by bounding best replies, as in Tadelis's Cournot example (§4.2.2, pp. 65–67) and item 3. Assume $a > c \geq 0$, $b > 0$, quantities $q_i \in [0, \infty)$, and write $m = (a - c)/(2b)$ for the monopoly output.

Solution. Payoff and best reply. With Q_{-i} the others' total output,

$$u_i = (a - b(q_i + Q_{-i}) - c)\, q_i = b\, q_i\, (2x - q_i), \qquad x = m - \tfrac{Q_{-i}}{2},$$

a concave parabola in q_i with peak at x. Hence $BR_i(Q_{-i}) = \max\{0,\, m - Q_{-i}/2\}$, and, as in item 3,

$$u_i(a', Q_{-i}) - u_i(t, Q_{-i}) = b\,(t - a')(t + a' - 2x). \tag{1}$$

So if every possible peak x lies in $[\underline{a}, \bar{a}]$, then outputs above $\bar{a}$ are strictly dominated by $\bar{a}$ and outputs below $\underline{a}$ by $\underline{a}$. An output that is the unique best reply to some surviving Q_{-i} is not dominated.

(a) Two firms. Here $Q_{-i} = q_j$. If $S_j^k = [L_k, U_k]$, the surviving best replies fill $[BR(U_k), BR(L_k)]$, so

$$L_{k+1} = \max\{0,\ m - \tfrac{U_k}{2}\}, \qquad U_{k+1} = m - \tfrac{L_k}{2}.$$

- Round 1: $q_j \in [0, \infty)$, so the best replies fill $[0, m]$ ($BR = m$ at $q_j = 0$, $BR = 0$ for $q_j \geq 2m$). Every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: $q_j \leq m$ gives $BR \geq m/2$. $S_i^2 = [m/2, m]$.
- Round 3: $S_i^3 = [m/2, 3m/4]$; round 4: $[5m/8, 3m/4]$; ...

From round 2 on, the bounds are interior and $U_{k+1} - L_{k+1} = \frac{1}{2}(U_k - L_k)$, so the width is $m/2^{k-1} \to 0$. Any limit point solves $q = m - q/2$, that is,

$$S_i^\infty = \left\{\tfrac{2m}{3}\right\} = \left\{\frac{a - c}{3b}\right\}, \qquad i = 1, 2.$$

IESDS predicts the Cournot outcome $q_1 = q_2 = (a - c)/(3b)$. (Tadelis's example, with $m = 45$, gives 30 in the same way.)

(b) Three firms. Now $Q_{-i} = q_j + q_h$.

- Round 1: $Q_{-i} \in [0, \infty)$, so the best replies fill $[0, m]$ and every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: now $Q_{-i} \in [0, 2m]$, and $BR(2m) = \max\{0, m - m\} = 0$. The best replies still fill all of $[0, m]$: each $q \in [0, m]$ is the unique best reply to $q_j = q_h = m - q \in [0, m]$, which survived round 1. Nothing is deleted: $S_i^2 = S_i^1$.

The process stops after one round with $S_i^\infty = [0, (a - c)/(2b)]$ for each firm. *No*, IESDS does not give a unique prediction with three firms. (The Cournot equilibrium $q_i = (a - c)/(4b)$, item 9 with $n = 3$ and $b = 1$, lies inside, but so does everything else in $[0, m]$.)

Interpretation. With two firms the best replies form a contraction: each round of "I know that you know ..." halves the range of plausible outputs. With three firms the rivals' *combined* output can be as large as $2m$, enough to make zero output a best reply, so the first-round bounds reproduce themselves and higher-order reasoning adds nothing. IESDS is weak in games with several players whose actions are strategic substitutes; Nash equilibrium (Lecture 2) still pins down the outcome.

Verification. `check_A1.py` verifies with sympy that the Cournot payoff equals $bq_i(2x - q_i)$ with $x = m - q_j/2$, the factorization (1), and that the fixed point of $q = m - q/2$ is $(a - c)/(3b)$. It runs the duopoly interval recursion for 60 rounds in exact arithmetic ($m = 1$: $[0, 1]$, $[\frac{1}{2}, 1]$, $[\frac{1}{2}, \frac{3}{4}]$, $\ldots$, $\to \frac{2}{3}$), checks that every $q \in [0, m]$ is the best reply to $Q = 2(m - q)$ with both rivals in $[0, m]$, and runs brute-force IESDS for three firms on a grid ($a - c = 12$, $b = 1$, steps of 0.5), which leaves exactly $[0, 6] = [0, m]$. *Comparison with the MWG solutions manual* (pp. 8-3–8-4): it agrees on both parts, $(a - c)/(3b)$ for two firms and $q_i \in [0, (a - c)/(2b)]$ with no unique prediction for three.

Answer. (a) $BR_i(q_j) = \max\{0, (a-c)/(2b) - q_j/2\}$; IESDS gives $[0, m]$, $[m/2, m]$, $[m/2, 3m/4]$, $\ldots$, with widths halving, so the unique survivor is $q_1 = q_2 = (a - c)/(3b)$. (b) No: with three firms the rivals' total can reach $2m$, every $q_i \in [0, (a-c)/(2b)]$ is a best reply to some surviving profile, and IESDS stops at $[0, (a - c)/(2b)]$ for each firm.

Lecture 2 — Static games II: Nash equilibrium and mixed strategies

7 Tadelis Exercise 5.6

Foundations · block L2

Source: Tadelis Exercise 5.6, page 96 (statement in the book and in Exercise 7 of the list; not reproduced here).

What is asked. (a) Write the Hawk-Dove game as a matrix for general v, $k > v/2$, $d < v/2$; (b) find the pure-strategy Nash equilibria for $v = 10$, $k = 6$, $d = 4$.

Method. Pure-strategy Nash equilibrium (Tadelis Def. 5.1, p. 80; Lecture 2, slide 14): a profile in which each player's strategy is a best response to the other's. Find it by marking best responses cell by cell (the underline method, Tadelis p. 81).

Solution. **Notation.** $v > 0$: value of the resource; k: injury cost when both fight, $k > v/2$; d: display cost when both display, $d < v/2$; $S_1 = S_2 = \{H, D\}$.

(a) The matrix. Read each cell from the story. Both H: they split v and each loses k, so each gets $v/2 - k$. Both D: they split v and each pays d, so each gets $v/2 - d$. One H, one D: the hawk gets v, the dove gets 0.

$1 \setminus 2$	H	D
H	$\frac{v}{2} - k,\ \frac{v}{2} - k$	$v,\ 0$
D	$0,\ v$	$\frac{v}{2} - d,\ \frac{v}{2} - d$

(b) Pure equilibria with $v = 10$, $k = 6$, $d = 4$. Substituting, $v/2 - k = 5 - 6 = -1$ and $v/2 - d = 5 - 4 = 1$:

$1 \setminus 2$	H	D
H	$-1,\ -1$	$10,\ 0$
D	$0,\ 10$	$1,\ 1$

Best responses of player 1 (the game is symmetric, so player 2's are the same):

- against H: $u_1(D,H) = 0 > -1 = u_1(H,H)$, so $BR_1(H) = \{D\}$;
- against D: $u_1(H,D) = 10 > 1 = u_1(D,D)$, so $BR_1(D) = \{H\}$.

Check the four cells:

- (H,H): each would switch to D ($0 > -1$). Not NE.
- (D,D): each would switch to H ($10 > 1$). Not NE.
- (H,D): $H \in BR_1(D)$ and $D \in BR_2(H)$. NE, payoffs $(10,0)$.
- (D,H): by symmetry, NE, payoffs $(0,10)$.

The same holds for every admissible (v,k,d). Against H, D is strictly better because $0 > v/2 - k \iff k > v/2$. Against D, H is strictly better because $v > v/2 - d \iff d > -v/2$, which holds whenever $d \geq 0$. So the two asymmetric profiles are the only pure equilibria under the book's assumptions; the numbers only make it concrete.

Interpretation. Hawk-Dove is an anti-coordination game (the same structure as Chicken, item 8): each player wants to do the opposite of the other. The two pure equilibria are asymmetric: one animal takes everything, the other backs off. Nothing in the symmetric game says *which* animal is the hawk, which is why the biology literature turns to a symmetric mixed (population) answer. Beyond what is asked: in the symmetric mixed equilibrium each plays H with probability p that makes the other indifferent, $-p + 10(1-p) = 0 \cdot p + 1 \cdot (1-p)$, so $p = 9/10$, with expected payoff $1/10$ each.

Verification. `check_A2.py` builds the matrix from the story with symbols, checks that the best responses are $BR(H) = D$, $BR(D) = H$ for random admissible (v,k,d), finds by brute force that the pure NE at $(10,6,4)$ are exactly (H,D) and (D,H), and checks the mixed $p = 9/10$. No answer in the book.

Answer. (a) (H,H): $(\frac{v}{2}-k, \frac{v}{2}-k)$; (H,D): $(v,0)$; (D,H): $(0,v)$; (D,D): $(\frac{v}{2}-d, \frac{v}{2}-d)$. (b) With $v=10$, $k=6$, $d=4$ the matrix is $\left(\begin{smallmatrix} -1,-1 & 10,0 \\ 0,10 & 1,1 \end{smallmatrix}\right)$; the pure-strategy Nash equilibria are (H,D) and (D,H), with payoffs $(10,0)$ and $(0,10)$.

8 Muñoz-Garcia Ch. 3, Exercise 1

Foundations · block L2

Source: Muñoz-Garcia Ch. 3, Exercise 1, page 86–87 (statement in the book and in Exercise 8 of the list; not reproduced here).

What is asked. For the Game of Chicken of Fig. 3.1: (a) strictly dominated strategies; (b) best responses; (c) pure-strategy Nash equilibria; (d) the mixed-strategy Nash equilibrium, with p = Pr(player 1 plays Straight), q = Pr(player 2 plays Straight); (e) graph both best-response functions and show (d) on it.

Method. Dominance (Lecture 1, slide 23); best responses and pure NE (Tadelis Defs. 4.7 and 5.1, pp. 72, 80; Lecture 2, slides 9–14); mixed NE through the principle of indifference (Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46): a player mixes only if the other's mix makes him indifferent.

Solution. Notation. St = Straight, Sw = Swerve. Matrix (Fig. 3.1): $(St, St) = (0, 0)$, $(St, Sw) = (3, 1)$, $(Sw, St) = (1, 3)$, $(Sw, Sw) = (2, 2)$. The game is symmetric.

(a) Player 1's rows over columns (St, Sw): $St = (0, 3)$, $Sw = (1, 2)$. Sw is better against St ($1 > 0$), St is better against Sw ($3 > 2$): neither row dominates the other. By symmetry the same holds for player 2. No strictly dominated strategy.

(b) $BR_1(St) = Sw$ (payoff $1 > 0$) and $BR_1(Sw) = St$ (payoff $3 > 2$). By symmetry $BR_2(St) = Sw$ and $BR_2(Sw) = St$. Each player wants the opposite of the other: an anti-coordination game.

(c) A cell is a pure NE when both strategies are best responses to each other. (St, Sw): $St = BR_1(Sw)$ and $Sw = BR_2(St)$, NE. (Sw, St): NE by symmetry. (St, St) and (Sw, Sw) are not, since in each both players would switch. The psNE are (St, Sw) with payoffs $(3, 1)$ and (Sw, St) with payoffs $(1, 3)$.

(d) Mixed equilibrium. Player 1's expected payoffs when player 2 plays St with probability q:

$$EU_1(St) = 0 \cdot q + 3(1 - q) = 3 - 3q, \tag{1}$$
$$EU_1(Sw) = 1 \cdot q + 2(1 - q) = 2 - q. \tag{2}$$

Player 1 is willing to mix only if (1) = (2):

$$3 - 3q = 2 - q \implies 1 = 2q \implies q^* = \tfrac{1}{2}.$$

Player 2's expected payoffs when player 1 plays St with probability p (her payoffs are 0 at (St, St), 3 at (Sw, St), 1 at (St, Sw), 2 at (Sw, Sw)):

$$EU_2(St) = 0 \cdot p + 3(1 - p) = 3 - 3p, \tag{3}$$
$$EU_2(Sw) = 1 \cdot p + 2(1 - p) = 2 - p. \tag{4}$$

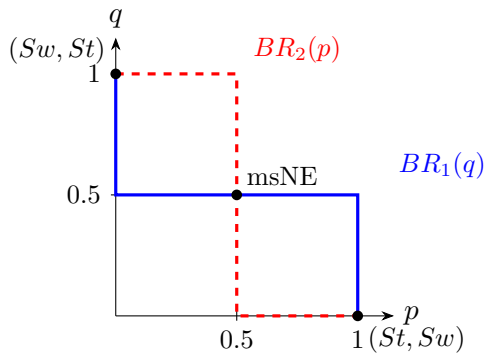
Setting (3) = (4): $3 - 3p = 2 - p$, so $p^* = \frac{1}{2}$. The msNE is

$$\sigma^* = \Big((\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw),\ (\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw) \Big),$$

and each player's expected payoff is $EU_1 = 3 - 3 \cdot \frac{1}{2} = 1.5$ (by (1) at $q^* = \frac{1}{2}$).

(e) Best-response functions. From (1)–(2), $EU_1(St) - EU_1(Sw) = 1 - 2q$, so

$$BR_1(q) = \begin{cases} p = 1 & q < \frac{1}{2}, \\ p \in [0, 1] & q = \frac{1}{2}, \\ p = 0 & q > \frac{1}{2}, \end{cases} \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{2}, \\ q \in [0, 1] & p = \frac{1}{2}, \\ q = 0 & p > \frac{1}{2}. \end{cases}$$



The two graphs cross three times: at $(p, q) = (1, 0)$, i.e. (St, Sw); at $(0, 1)$, i.e. (Sw, St); and at $(\frac{1}{2}, \frac{1}{2})$, the msNE.

Interpretation. Chicken has the shape of Hawk-Dove (item 7). In the mixed equilibrium each driver goes straight half of the time, so they crash with probability $\frac{1}{4}$, and each expects 1.5, less than the 2 of mutual swerving and less than the 3 of the favourable pure equilibrium. The mix is pinned by the *other* player's indifference, not by one's own payoffs (slide 50's lesson).

Verification. `check_A2.py` checks (a)–(c) by brute force, solves both indifference conditions with sympy ($q^* = p^* = \frac{1}{2}$, payoff $\frac{3}{2}$), and finds on a grid of $(p, q) \in \{0, \frac{1}{100}, \dots, 1\}^2$ that the mutual best responses are exactly $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$. *Comparison with the book's answer* (pp. 87–91): it agrees on all five parts (no dominated strategies, the same best responses, the two psNE, msNE with $p = q = \frac{1}{2}$, the same best-response graph, Figs. 3.4–3.6). One typo in the book: on p. 89 the rearranged indifference condition is printed "$3 + 3q = q + 2 - 2q \Rightarrow q = 1/2$"; from $0q + 3(1 - q) = 1q + 2(1 - q)$ it should read $3 - 3q = q + 2 - 2q$ (as printed, $3 + 3q = 2 - q$ gives $q = -\frac{1}{4}$). The book's result $q = \frac{1}{2}$ is correct.

Answer. (a) No strictly dominated strategy for either player. (b) $BR_i(St) = Sw$, $BR_i(Sw) = St$. (c) psNE: (St, Sw) and (Sw, St). (d) msNE: each plays Straight with probability $\frac{1}{2}$ ($p^* = q^* = \frac{1}{2}$), expected payoff 1.5 each. (e) $BR_1(q)$: $p = 1$ for $q < \frac{1}{2}$, $p = 0$ for $q > \frac{1}{2}$, any p at $q = \frac{1}{2}$ (symmetric for player 2); the graphs cross at $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$.

9 Tadelis Exercise 5.8

Exam · block L2

Source: Tadelis Exercise 5.8, page 96–97 (statement in the book and in Exercise 9 of the list; not reproduced here).

What is asked. Cournot with n identical firms, $P(Q) = a - Q$, unit cost $c < a$: (a) write the normal form; (b) find the Nash (Cournot) equilibrium; (c) find the limit of the equilibrium price as $n \to \infty$ and say what it is.

Method. Nash equilibrium with continuous strategies as the crossing of best responses (Tadelis §5.2.3, pp. 87–88; Lecture 2, slides 20–27, with the n-firm case on slides 26–27).

Solution. Book typo. The statement prints "$Q = q_i + \cdots + q_n$". Aggregate production is the sum over all firms, $Q = q_1 + \cdots + q_n$; the subscript i is a typo for 1, and the solution uses $Q = \sum_{j=1}^{n} q_j$.

Notation. $N = \{1, \dots, n\}$; $q_i \geq 0$ firm i's output; $Q = \sum_j q_j$; $Q_{-i} = Q - q_i$ the rivals' total; $a > c \geq 0$. The book defines $P(Q) = a - Q$ for $Q \leq a$; for $Q > a$ we take $P = 0$ (Gibbons's convention, p. 15). This only matters off equilibrium: there a firm earns $-c\,q_i \leq 0$.

(a) Normal form. Players N; strategies $S_i = [0, \infty)$; payoffs

$$u_i(q_i, q_{-i}) = \max\{0,\ a - q_i - Q_{-i}\}\, q_i - c\, q_i. \tag{1}$$

(b) Best response. Fix Q_{-i}. Raising q_i beyond $a - Q_{-i}$ only adds cost, so consider $q_i \leq a - Q_{-i}$, where $u_i = (a - c - Q_{-i} - q_i)\, q_i$ is strictly concave ($\partial^2 u_i / \partial q_i^2 = -2$). The first-order condition is

$$a - c - Q_{-i} - 2q_i = 0, \tag{2}$$

and with the constraint $q_i \geq 0$,

$$BR_i(Q_{-i}) = \max\Big\{0,\ \frac{a - c - Q_{-i}}{2}\Big\}. \tag{3}$$

Every firm produces in equilibrium. Suppose some firm has $q_i^* = 0$. By (3) that requires $Q_{-i}^* \geq a - c$, so some rival j produces $q_j^* > 0$ and, by (2) for j, $q_j^* = a - c - Q^*$ (because $Q_{-j}^* + 2q_j^* = Q^* + q_j^*$). But $Q^* \geq Q_{-i}^* \geq a - c$ makes $q_j^* \leq 0$, a contradiction. So every firm is interior and (2) holds for all i.

Solving. Rewrite (2) with $Q_{-i} = Q - q_i$:

$$a - c - (Q - q_i) - 2q_i = 0 \implies q_i = a - c - Q. \tag{4}$$

The right side of (4) is the same for every firm, so all firms produce the same amount, $q_i = q$. Summing (4) over the n firms:

$$Q = n(a - c - Q) \implies (n+1)Q = n(a-c) \implies Q^* = \frac{n(a-c)}{n+1}. \tag{5}$$

Substituting (5) into (4):

$$q_i^* = a - c - \frac{n(a-c)}{n+1} = \frac{a-c}{n+1}, \qquad i = 1, \ldots, n. \tag{6}$$

Since $Q^* < a - c < a$, the price is on the linear part of demand:

$$P^* = a - Q^* = a - \frac{n(a-c)}{n+1} = \frac{a+nc}{n+1}, \tag{7}$$

and each firm earns

$$u_i^* = (P^* - c)\, q_i^* = \Big(\frac{a+nc}{n+1} - c\Big)\frac{a-c}{n+1} = \Big(\frac{a-c}{n+1}\Big)^2.$$

The equilibrium is unique: the argument above showed every NE is interior, and (4)–(6) have one solution. In the slides' notation, $q^m = (a-c)/2$ and (6) reads $q_i^* = 2q^m/(n+1)$; $n = 1$ gives the monopoly output q^m and $n = 2$ gives $\frac{2}{3}q^m$ (slide 24).

(c) Limit. From (7),

$$P^* = \frac{a+nc}{n+1} = c + \frac{a-c}{n+1} \xrightarrow[n\to\infty]{} c,$$

and $Q^* \to a - c = q^{cp}$, the competitive output, with each firm's output and profit going to zero. This is the familiar perfectly competitive outcome: price equals marginal cost.

Interpretation. The markup $P^* - c = (a-c)/(n+1)$ falls as $1/(n+1)$. Each firm's output lowers the price for everyone, but a firm counts only the loss on its own units; with more firms its own share is smaller, so it expands more aggressively and the outcome approaches competition (slide 27). Contrast with Bertrand (Lecture 2, slides 29–40), where two firms already price at cost.

Verification. `check_A2.py` solves the n first-order conditions with sympy for $n = 1, \ldots, 6$ and checks (5)–(7) and the profit formula, checks that $P^* - c = (a-c)/(n+1)$ and that the limit is c, and confirms numerically ($a = 100$, $c = 10$, $n = 3$) that no unilateral deviation on a fine grid beats $q_i^* = 22.5$, including the kinked part of demand. No answer in the book.

Answer. (a) $N = \{1, \ldots, n\}$, $S_i = [0, \infty)$, $u_i = (a - \sum_j q_j)\, q_i - c\, q_i$ (with price 0 if $\sum_j q_j > a$); the printed "$Q = q_i + \cdots + q_n$" should be $q_1 + \cdots + q_n$. (b) Unique NE $q_i^* = \frac{a-c}{n+1}$, $Q^* = \frac{n(a-c)}{n+1}$, $P^* = \frac{a+nc}{n+1}$, profit $\big(\frac{a-c}{n+1}\big)^2$. (c) $P^* \to c$: the perfectly competitive price (price = marginal cost).

10 Tadelis Exercise 6.8

Exam · block L2

Source: Tadelis Exercise 6.8, page 125 (statement in the book and in Exercise 10 of the list; not reproduced here).

What is asked. Three firms decide whether to enter a market; an entrant earns $150/n - 62$ when n firms enter, a non-entrant 0. (a) Find all pure-strategy NE; (b) find the symmetric mixed-strategy NE in which every firm enters with the same probability.

Method. Pure NE by checking unilateral deviations (Tadelis Def. 5.1, p. 80); mixed NE by the principle of indifference (Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46), here with three players.

Solution. **Notation.** $I = \{1, 2, 3\}$, $S_i = \{E, O\}$ (enter, stay out). With n entrants an entrant's payoff is $\pi(n) = 150/n - 62$; staying out pays 0:

$$\pi(1) = 150 - 62 = 88, \qquad \pi(2) = 75 - 62 = 13, \qquad \pi(3) = 50 - 62 = -12.$$

(a) Pure equilibria. Classify profiles by the number n of entrants.

- $n = 0$: any firm that enters gets $\pi(1) = 88 > 0$. Not NE.
- $n = 1$: an outsider that enters gets $\pi(2) = 13 > 0$. Not NE.
- $n = 2$: each entrant gets $13 > 0$, so it does not want to leave; the outsider would get $\pi(3) = -12 < 0$ by entering, so it stays out. NE.
- $n = 3$: each gets $-12 < 0$ and would rather stay out. Not NE.

The pure NE are the three profiles in which exactly two firms enter: (E, E, O), (E, O, E), (O, E, E), each with payoffs 13 to the entrants and 0 to the outsider.

(b) Symmetric mixed equilibrium. Let each firm enter with probability p. The pure symmetric profiles are not NE (by (a), $p = 0$ and $p = 1$ fail), so $p \in (0, 1)$ and every firm must be indifferent between E and O. Fix firm i. The number of other entrants is 0, 1 or 2 with probabilities $(1-p)^2$, $2p(1-p)$, p^2, so

$$u_i(E, p) = (1-p)^2 \cdot 88 + 2p(1-p) \cdot 13 + p^2 \cdot (-12), \qquad u_i(O, p) = 0. \tag{1}$$

Expanding the first expression in (1):

$$88(1 - 2p + p^2) + 26(p - p^2) - 12p^2 = 88 - 176p + 88p^2 + 26p - 26p^2 - 12p^2 = 88 - 150p + 50p^2.$$

Indifference, $u_i(E, p) = 0$, gives

$$50p^2 - 150p + 88 = 0 \iff 25p^2 - 75p + 44 = 0,$$

$$p = \frac{75 \pm \sqrt{75^2 - 4 \cdot 25 \cdot 44}}{2 \cdot 25} = \frac{75 \pm \sqrt{5625 - 4400}}{50} = \frac{75 \pm 35}{50} \in \left\{\frac{4}{5}, \frac{11}{5}\right\}.$$

Only $p = 4/5$ is a probability. The symmetric mixed NE has each firm entering with probability $p^* = \frac{4}{5}$, and each firm's expected payoff is 0 (the payoff of staying out).

Check: at $p = 0.8$, $(0.2)^2 \cdot 88 + 2(0.8)(0.2) \cdot 13 + (0.8)^2 \cdot (-12) = 3.52 + 4.16 - 7.68 = 0$.

Interpretation. Entry is a coordination problem over *who* enters: the market supports two firms, not three. The pure equilibria solve it but need the firms to know who stays out. Without such a convention, the symmetric equilibrium has each firm entering with probability 0.8. Then all three enter with probability $0.8^3 = 0.512$ and lose money, and the expected profit of the industry is zero: all the rents of the two-firm market are dissipated by miscoordination.

Verification. `check_A2.py` brute-forces the 2^3 profiles (pure NE are exactly the three two-entrant profiles), solves the indifference condition with sympy (roots $\frac{4}{5}$ and $\frac{11}{5}$), and checks that at $p = \frac{4}{5}$ entering and staying out both pay 0 while $u_i(E, p)$ is positive for $p < \frac{4}{5}$ and negative for $p \in (\frac{4}{5}, 1]$. No answer in the book.

Answer. (a) The three profiles with exactly two entrants, (E, E, O), (E, O, E), (O, E, E): each entrant earns $75 - 62 = 13$, and the outsider would earn $50 - 62 = -12$ by entering. (b) Each firm enters with probability $p^* = \frac{4}{5}$ (from $50p^2 - 150p + 88 = 0$); expected payoff 0.

11 Gibbons Problem 1.6

Exam · block L2

Source: Gibbons Problem 1.6, page 49 (statement in the book and in Exercise 11 of the list; not reproduced here).

What is asked. Cournot duopoly with $P(Q) = a - Q$ and marginal costs c_1, c_2: find the NE when $0 < c_i < a/2$ for both firms, and when $c_1 < c_2 < a$ but $2c_2 > a + c_1$.

Method. Cournot equilibrium as the crossing of best responses (Gibbons §1.2.A, pp. 14–21; Tadelis §5.2.3, eq. (5.2), p. 87; Lecture 2, slides 20–24), now with a corner: a best response can be $q_i = 0$.

Solution. **Notation.** $q_i \geq 0$; $P(Q) = a - Q$ for $Q < a$ and 0 for $Q \geq a$ (the referenced model, Gibbons p. 15); firm i's cost $c_i q_i$, $0 \leq c_i < a$; $j \neq i$.

Best responses. For $q_i < a - q_j$ the payoff is

$$u_i(q_i, q_j) = (a - q_i - q_j - c_i)\, q_i, \tag{1}$$

strictly concave in q_i; for $q_i \geq a - q_j$ the price is zero and $u_i = -c_i q_i \leq 0$. The first-order condition of (1) is $a - c_i - q_j - 2q_i = 0$, so

$$BR_i(q_j) = \max\Big\{0,\ \frac{a - c_i - q_j}{2}\Big\}. \tag{2}$$

The corner $BR_i = 0$ occurs exactly when $q_j \geq a - c_i$: the rival alone already pushes the price down to c_i or below.

Interior candidate. If both are interior, (2) gives $2q_1 = a - c_1 - q_2$ and $2q_2 = a - c_2 - q_1$. Substituting $q_2 = (a - c_2 - q_1)/2$ into the first:

$$2q_1 = a - c_1 - \Big(\frac{a - c_2 - q_1}{2}\Big) \implies 4q_1 = 2a - 2c_1 - a + c_2 + q_1 \implies q_1^* = \frac{a - 2c_1 + c_2}{3}, \tag{3}$$

and symmetrically

$$q_2^* = \frac{a - 2c_2 + c_1}{3}. \tag{4}$$

This is a NE exactly when both (3) and (4) are positive.

Case 1: $0 < c_i < a/2$ **for** $i = 1, 2$**.** Then $a - 2c_i > 0$, so $a - 2c_i + c_j > c_j > 0$: (3) and (4) are positive and form a NE. No other NE exists. A corner NE with $q_2 = 0$ would need $q_1 = BR_1(0) = (a - c_1)/2$ and, by (2), $q_1 \geq a - c_2$, that is, $(a - c_1)/2 \geq a - c_2 \iff 2c_2 \geq a + c_1$, impossible because $2c_2 < a < a + c_1$. The same holds for $q_1 = 0$, and $(0, 0)$ fails because $BR_i(0) = (a - c_i)/2 > 0$. The unique NE is

$$q_1^* = \frac{a - 2c_1 + c_2}{3}, \quad q_2^* = \frac{a - 2c_2 + c_1}{3}, \quad Q^* = \frac{2a - c_1 - c_2}{3}, \quad P^* = \frac{a + c_1 + c_2}{3},$$

with profits $u_i^* = (P^* - c_i)\, q_i^* = (q_i^*)^2$, because $P^* - c_i = (a + c_1 + c_2 - 3c_i)/3 = (a - 2c_i + c_j)/3 = q_i^*$.

Case 2: $c_1 < c_2 < a$ **and** $2c_2 > a + c_1$**.** Now (4) is negative, $a - 2c_2 + c_1 < 0$, so there is no interior NE. Try firm 1 as a monopolist:

$$q_1 = BR_1(0) = \frac{a - c_1}{2}, \qquad q_2 = 0.$$

Firm 2's best response to it, by (2):

$$BR_2\Big(\frac{a - c_1}{2}\Big) = \max\Big\{0,\ \frac{a - c_2 - \frac{a - c_1}{2}}{2}\Big\} = \max\Big\{0,\ \frac{a + c_1 - 2c_2}{4}\Big\} = 0,$$

since $a + c_1 - 2c_2 < 0$. So $(\frac{a - c_1}{2}, 0)$ is a NE. It is the only one: the other corner, $q_1 = 0$ and $q_2 = (a - c_2)/2$, would need $(a - c_2)/2 \geq a - c_1$, that is, $2c_1 \geq a + c_2$, but $2c_1 < c_1 + c_2 < a + c_2$; and $(0, 0)$ fails as before. At the NE the price is the monopoly price of firm 1,

$$P^* = a - \frac{a - c_1}{2} = \frac{a + c_1}{2} < c_2,$$

so firm 2 cannot sell any unit profitably; firm 1 earns $\big(\frac{a - c_1}{2}\big)^2$.

Interpretation. With similar costs both firms produce, and the cheaper one produces more: $q_1^* - q_2^* = c_2 - c_1$. When firm 2's cost exceeds firm 1's monopoly price, $c_2 > (a + c_1)/2$, firm 2 is priced out even though it is not shut out by any rule: firm 1 simply behaves as a monopolist and firm 2's best reply is to stay out. The condition $2c_2 > a + c_1$ is exactly "c_2 above firm 1's monopoly price". (Case 1 is sufficient, not necessary, for the interior NE: (3)–(4) are positive whenever $2c_i < a + c_j$ for both firms, even if some $c_i \geq a/2$.)

Verification. `check_A2.py` derives (2)–(4) with sympy, checks $u_i^* = (q_i^*)^2$, and for random cost pairs in each case finds the NE by iterating the exact best responses (2) and confirms it equals the formulas above; it also checks on a fine grid that no unilateral deviation pays, including outputs that push Q above a. No answer in the book. (Kumar's unofficial Gibbons manual, not in `Livros/`, was not consulted.)

Answer. If $0 < c_i < a/2$: unique NE $q_1^* = \frac{a - 2c_1 + c_2}{3}$, $q_2^* = \frac{a - 2c_2 + c_1}{3}$, price $\frac{a + c_1 + c_2}{3}$, profits $(q_i^*)^2$. If $c_1 < c_2 < a$ and $2c_2 > a + c_1$: unique NE $q_1^* = \frac{a - c_1}{2}$, $q_2^* = 0$; firm 1 is a monopolist at price $\frac{a + c_1}{2} < c_2$.

12 MWG Exercise 8.D.6

Advanced · block L2

Source: MWG Exercise 8.D.6, page 264 (statement in the book and in Exercise 12 of the list; not reproduced here).

What is asked. Prove that every two-player game with two strategies each, with arbitrary payoffs $u, v, \ell, m, w, x, y, z$ as in the list's matrix, has a mixed-strategy Nash equilibrium.

Method. The hint's route: describe each player's strategy by one probability, write the best-response correspondences, and show they always intersect (MWG §8.D, Def. 8.D.2 and Prop. 8.D.1, pp. 250–251, which is the slides' principle of indifference, Lecture 2, slides 44–46; Tadelis §6.4, pp. 121–122; Gibbons §1.3.B, pp. 41–44). Pure equilibria count: they are degenerate mixed strategies. Nash's theorem (slide 51) gives the result in general; the point here is a direct proof.

Solution. Notation. $p = \sigma_1(a_1) \in [0,1]$, $q = \sigma_2(b_1) \in [0,1]$. Payoffs: $u_1(a_1,b_1) = u$, $u_1(a_1,b_2) = \ell$, $u_1(a_2,b_1) = w$, $u_1(a_2,b_2) = y$; $u_2(a_1,b_1) = v$, $u_2(a_1,b_2) = m$, $u_2(a_2,b_1) = x$, $u_2(a_2,b_2) = z$.

Step 1: the payoff gains are affine. Player 1's gain from a_1 over a_2 when player 2 plays q:

$$D_1(q) = [qu + (1-q)\ell] - [qw + (1-q)y] = q(u-w) + (1-q)(\ell - y). \tag{1}$$

Player 2's gain from b_1 over b_2 when player 1 plays p:

$$D_2(p) = [pv + (1-p)x] - [pm + (1-p)z] = p(v-m) + (1-p)(x-z). \tag{2}$$

Player 1's expected payoff is $u_1(p,q) = u_1(a_2,q) + p\,D_1(q)$, linear in p, so

$$BR_1(q) = \begin{cases} \{1\} & D_1(q) > 0, \\ [0,1] & D_1(q) = 0, \\ \{0\} & D_1(q) < 0, \end{cases} \qquad BR_2(p) = \begin{cases} \{1\} & D_2(p) > 0, \\ [0,1] & D_2(p) = 0, \\ \{0\} & D_2(p) < 0. \end{cases} \tag{3}$$

(p^*, q^*) is a NE iff $p^* \in BR_1(q^*)$ and $q^* \in BR_2(p^*)$.

Step 2: a sign lemma. An affine function D on $[0,1]$ satisfies exactly one of: (i) $D(t) \geq 0$ for all $t \in [0,1]$; (ii) $D(t) \leq 0$ for all t, with $D \not\equiv 0$; or (iii) $D(0)$ and $D(1)$ are nonzero with opposite signs. In case (iii) D has a unique zero $t^* = D(0)/(D(0) - D(1)) \in (0,1)$. (If $D(0)$, $D(1)$ are both ≥ 0 or both ≤ 0, then D, a convex combination of them, keeps that sign on $[0,1]$.)

Step 3: the proof, by cases on D_1.

Case A: $D_1(q) \geq 0$ for all q. Then $p = 1 \in BR_1(q)$ for every q by (3). Let $\hat{q} \in BR_2(1)$; it exists because (3) never gives an empty set. Then $1 \in BR_1(\hat{q})$ and $\hat{q} \in BR_2(1)$: $(1, \hat{q})$ is a NE.

Case B: $D_1(q) \leq 0$ for all q. Symmetrically, $p = 0$ is a best reply to every q, and $(0, \hat{q})$ with $\hat{q} \in BR_2(0)$ is a NE.

Case C: $D_1(0)$ and $D_1(1)$ have opposite signs. Player 1 is indifferent at exactly one $q^* \in (0,1)$. Solving $D_1(q^*) = 0$ from (1):

$$q^* = \frac{y - \ell}{(u-w) + (y-\ell)}. \tag{4}$$

Now look at D_2 and apply the lemma again.

- If $D_2(p) \geq 0$ for all p: $q = 1$ is a best reply to every p; with $\hat{p} \in BR_1(1)$, $(\hat{p}, 1)$ is a NE.
- If $D_2(p) \leq 0$ for all p: with $\hat{p} \in BR_1(0)$, $(\hat{p}, 0)$ is a NE.
- If $D_2(0)$, $D_2(1)$ have opposite signs: player 2 is indifferent at

$$p^* = \frac{z - x}{(v-m) + (z-x)} \in (0,1). \tag{5}$$

  At (p^*, q^*), $D_1(q^*) = 0$ gives $BR_1(q^*) = [0,1] \ni p^*$, and $D_2(p^*) = 0$ gives $BR_2(p^*) = [0,1] \ni q^*$. So (p^*, q^*) is a NE, in strictly mixed strategies.

The cases A, B, C exhaust all possibilities by the lemma, and each produces a NE. Hence every game of this form has a mixed-strategy Nash equilibrium. ■

Interpretation. The argument is the 2×2 version of the fixed-point proof (Tadelis pp. 121–122): each best-response correspondence is a "step" from one side of the unit square to the other, with a vertical (or horizontal) segment where the player is indifferent, and two such steps must cross. Mixing is what makes this work: in pure strategies the steps have gaps (Matching Pennies has no pure NE), and the indifference segment closes the gap. Gibbons's four cases (p. 42) are the sign patterns of $D_1(0) = \ell - y$ and $D_1(1) = u - w$.

Verification. `check_A2.py` checks (1), (2), (4) and (5) with sympy, then draws 20,000 random 2×2 games (half with integer payoffs in $\{-2, \dots, 2\}$, so that ties and constant gains occur often), runs the case construction above, and confirms with the exact best-response test (3) that the profile it returns is a NE every time. *Comparison with the MWG solutions manual* (pp. 8-15–8-16, where the OCR in `Livros/` prints the number as "8.0.6"): it splits into the four strict-dominance cases ($u > w,\ \ell > y$; $u < w,\ \ell < y$; $v > m,\ x > z$; $v < m,\ x < z$) plus "all other cases", where it gives the mixed equilibrium $\alpha = (y-\ell)/(u+y-\ell-w)$ for player 2 and $\beta = (z-x)/(v+z-x-m)$ for player 1. These are our (4) and (5); the conclusion agrees. Two remarks on the manual: (i) in cases 3 and 4 it says "Player 2 will play his best response" where it means player 1 (a slip: the strategies named are a_1, a_2); (ii) its "all other cases" is not complete as written: if $u = w$ and $\ell = y$ (player 1 indifferent to everything) and player 2 has no strictly dominant strategy, the formula for α is $0/0$ (any α then works, so an equilibrium still exists, but the manual's formula does not exhibit it); the same holds for β when $v = m$ and $x = z$. The case split by the sign of D_1 above has no such gap.

Answer. With $p = \Pr(a_1)$, $q = \Pr(b_1)$, player 1's gain from a_1 is $D_1(q) = q(u-w)+(1-q)(\ell-y)$ and player 2's gain from b_1 is $D_2(p) = p(v-m) + (1-p)(x-z)$, both affine. If $D_1 \geq 0$ (or ≤ 0) on $[0,1]$, player 1 has a pure strategy that is always a best reply, and pairing it with player 2's best reply to it is a NE; the same if D_2 keeps a sign. Otherwise both change sign, at $q^* = \frac{y-\ell}{(u-w)+(y-\ell)}$ and $p^* = \frac{z-x}{(v-m)+(z-x)}$ in $(0,1)$, and (p^*, q^*) is a NE because each player is indifferent. So a NE always exists.

Lecture 3 — Dynamic games and subgame perfection

13 Tadelis Exercise 8.8

Foundations · block L3

Source: Tadelis Exercise 8.8, page 171–172 (statement in the book and in Exercise 13 of the list; not reproduced here).

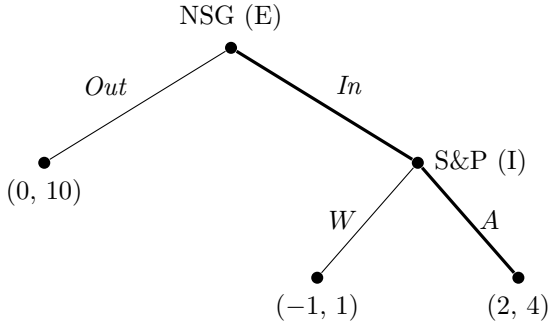
What is asked. Set up the game tree of the entry story (NSG enters or not, then S&P fights or accommodates) and find the subgame-perfect equilibrium outcome.

Method. Backward induction in a finite game of perfect information (Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 151–153). With perfect information the subgame-perfect equilibria are exactly the profiles that survive backward induction (Tadelis §8.2, p. 158). Same structure as the entry game of Cabral §7.2 (`Notas/L03/L03-06`).

Solution. **Notation.** Players: NSG (the entrant, E) and S&P (the incumbent, I). E chooses *In* or *Out*; after *In*, I chooses W (price war) or A (accommodate). Payoffs (u_E, u_I) in \$ million of

net present value. The entry cost of 2 is sunk once E enters, so it is subtracted from every payoff that follows *In*:

$$(u_E, u_I)(Out) = (0,\ 10),$$
$$(u_E, u_I)(In, W) = (1-2,\ 1) = (-1,\ 1),$$
$$(u_E, u_I)(In, A) = (4-2,\ 4) = (2,\ 4).$$



Payoffs (u_E, u_I); thick branches = backward-induction choices.

Step 1: the incumbent's node. After entry, S&P compares

$$u_I(In, A) = 4 > 1 = u_I(In, W),$$

so it accommodates. The price war is not a credible response.

Step 2: the entrant's node. Replacing S&P's node by its outcome $(2, 4)$, NSG compares

$$u_E(In) = 2 > 0 = u_E(Out),$$

so it enters.

Step 3: the non-credible Nash equilibrium. In the normal form (rows: E's strategies; columns: I's strategies)

	W	A
Out	0, 10	0, 10
In	−1, 1	2, 4

there are two pure Nash equilibria, (In, A) and (Out, W). In (Out, W) the war is never fought, so the threat costs S&P nothing; it fails sequential rationality at the node S&P never reaches (Tadelis Def. 7.8, p. 146; §8.1).

Interpretation. S&P's prediction of a price war is cheap talk: once NSG has sunk its \$2 million, a war gives S&P 1 instead of 4, and NSG knows it. The incumbent would have to change its own payoffs (capacity, a contract, a bond) to make the war credible, as in Cabral §7.2. The outcome does not depend on how the entry cost is booked: if "each firm earns \$1 million" were already net of it, NSG would still enter ($2 > 0$, and even a war would pay $1 > 0$).

Verification. `check_B_L03.py` recomputes both backward-induction steps and the two pure Nash equilibria of the 2×2 normal form. Tadelis prints no answer.

Answer. Tree above. Subgame-perfect equilibrium: NSG plays *In*, S&P plays *A* (accommodate); outcome: entry and accommodation, payoffs $(2, 4)$ in \$ million (NSG net of the entry cost). (Out, W) is a Nash equilibrium supported by a non-credible threat.

14 Muñoz-Garcia Ch. 4 Exercise 3

Foundations · block L3

Source: Muñoz-Garcia Ch. 4 Exercise 3, page 160 (statement in the book and in Exercise 14 of the list; not reproduced here).

What is asked. In the sequential prisoner's dilemma of Fig. 4.9: (a) player 2's best reply to each move of player 1; (b) player 1's move and the subgame-perfect equilibrium; (c) whether play differs from the simultaneous version (Ch. 1, Fig. 1.8).

Method. Backward induction (Tadelis §8.1, Prop. 8.1, p. 153). An SPE is written as complete strategies: player 2's strategy names a reply at *both* of her nodes, including the one off the path (Tadelis Def. 7.4, p. 139; Gibbons pp. 125–126 on outcome vs. equilibrium).

Solution. **Notation.** C = Confess, N = Not confess. Payoffs (u_1, u_2) from Fig. 4.9: $(C,C) \mapsto (-5,-5)$, $(C,N) \mapsto (0,-15)$, $(N,C) \mapsto (-15,0)$, $(N,N) \mapsto (-1,-1)$. Player 2's strategy is a pair $s_2 = (s_2(C), s_2(N))$: her reply after C, then after N. So $S_2 = \{CC, CN, NC, NN\}$ and $S_1 = \{C, N\}$.

(a) Player 2's best replies.

$$\begin{aligned} &\text{after } C: \quad u_2(C,C) = -5 > -15 = u_2(C,N) \;\Rightarrow\; C; \\ &\text{after } N: \quad u_2(N,C) = 0 > -1 = u_2(N,N) \;\Rightarrow\; C. \end{aligned}$$

So $s_2^* = CC$: confess whatever player 1 did.

(b) Player 1's move. Replacing each of player 2's nodes by its outcome,

$$u_1(C, s_2^*) = u_1(C,C) = -5 \;>\; -15 = u_1(N,C) = u_1(N, s_2^*),$$

so player 1 confesses. The SPE is

$$(s_1^*, s_2^*) = \big(C,\ (C \text{ after } C,\ C \text{ after } N)\big) = (C, CC),$$

with outcome (C,C) and payoffs $(-5,-5)$. In the normal form (2×4) this is also the only pure Nash equilibrium. If player 1 plays C, player 2 must reply C after C (CC or CN), and against CN player 1 would switch to N ($-1 > -5$). If player 1 plays N, player 2 must reply C after N (CC or NC), and then C pays player 1 -5 or 0, more than -15.

(c) Comparison. In the simultaneous game (Fig. 1.8) C is strictly dominant for each prisoner ($-5 > -15$ and $0 > -1$), so the unique Nash equilibrium is (C,C) with $(-5,-5)$. The sequential version has the same outcome. Seeing player 1's move does not change player 2's choice, because C is her best reply to each of player 1's moves; player 1 therefore gains nothing from moving first.

Interpretation. Observing the other's move matters only when the best reply depends on it. Here it does not: confessing is dominant at both of player 2's nodes, so the sequential game unravels exactly as the simultaneous one, and the Pareto-better (N,N) is still out of reach. Making a move observable is a commitment device only when it changes what the follower wants to do (item 17, Stackelberg).

Verification. `check_B_L03.py` computes player 2's replies, player 1's choice, the pure Nash equilibria of the 2×4 normal form (only (C, CC)) and of the simultaneous game (only (C,C)). *Comparison with Muñoz-Garcia's answer* (pp. 160–162): it agrees on all three parts, SPNE =

{(Confess, Confess, Confess)}. One wording slip there: the third component is described as player 2's response "to player 1 selecting Confess (in Subgame 3)"; Subgame 3 follows Not Confess, so it is the reply to Not Confess.

Answer. (a) Player 2 confesses after Confess ($-5 > -15$) and after Not confess ($0 > -1$). (b) Player 1 confesses ($-5 > -15$); SPE $= (C, (C, C))$, outcome (C, C), payoffs $(-5, -5)$. (c) No difference: the simultaneous game's unique equilibrium is also (C, C), since Confess is strictly dominant at every node.

15 Tadelis Exercise 8.10

Exam · block L3

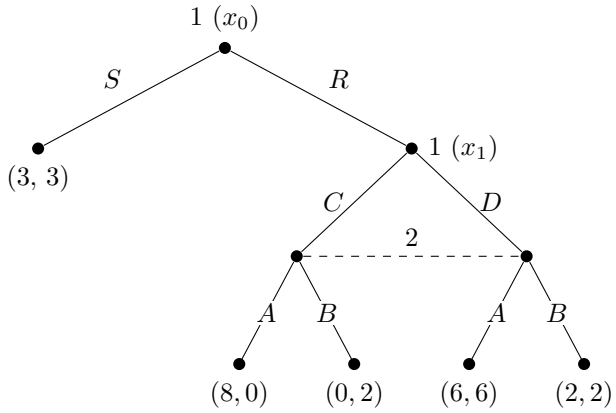
Source: Tadelis Exercise 8.10, page 172 (statement in the book and in Exercise 15 of the list; not reproduced here).

What is asked. Player 1 either plays safe (S, payoffs $(3, 3)$) or risks (R) a simultaneous 2×2 game with player 2. (a) Draw a tree and count its proper subgames; (b) say whether other trees would do; (c) write the normal form; (d) find all Nash and subgame-perfect equilibria.

Method. Extensive form with an information set for the simultaneous stage (Tadelis §7.1, Def. 7.2, p. 135); proper subgames (Def. 8.2, pp. 154–155; Tadelis counts the whole game); strategies as complete plans (Def. 7.4, p. 139); SPE by replacing the subgame with each of its Nash equilibria (§8.2, Def. 8.3, p. 157; the voluntary Battle of the Sexes, pp. 154–159, has the same structure). Mixed equilibria by the indifference principle (Tadelis §6.1–6.2, pp. 101–114).

Solution. **Notation.** After R, player 1 chooses C or D and player 2 chooses A or B without seeing it. $p = \Pr(C)$, $q = \Pr(A)$ in the subgame.

(a) Tree and subgames.



Player 2's two nodes form one information set (dashed). Candidate roots of a proper subgame are the singleton nodes x_0 and x_1; player 2's nodes fail (each would cut her information set). So the game has **two proper subgames**: the whole game and the simultaneous game that starts at x_1 (one, if, like Gibbons, one does not count the whole game).

(b) Other trees. Yes. (i) In the simultaneous stage, player 2 can be drawn as moving first at x_1 with player 1's two nodes joined in an information set; the strategies, normal form and subgames are unchanged. (ii) A different tree also represents the story: player 1 picks S, RC or RD at the root, and player 2 has one information set over the nodes after RC and RD. Its normal form is

the same up to merging SC and SD (it is the *reduced* normal form), but it has only one proper subgame, the whole game, so subgame perfection there has no bite: every Nash equilibrium is subgame perfect. The choice of tree matters for SPE whenever it changes which subgames exist; tree (i) and the drawing in (a) are the faithful ones, since R is a real decision taken before the stage game.

(c) Normal form. Player 1 has two information sets, so $S_1 = \{SC, SD, RC, RD\}$ (first letter at x_0, second at x_1); $S_2 = \{A, B\}$:

	A	B
SC	3, 3	3, 3
SD	3, 3	3, 3
RC	8, 0	0, 2
RD	6, 6	2, 2

(d) Equilibria.

Step 1: the subgame at x_1. Best replies: $BR_1(A) = C$ ($8 > 6$), $BR_1(B) = D$ ($2 > 0$), $BR_2(C) = B$ ($2 > 0$), $BR_2(D) = A$ ($6 > 2$). They cycle, so there is no pure equilibrium. Player 2 is indifferent when

$$u_2(A) = 0 \cdot p + 6(1-p) \;=\; 2p + 2(1-p) = u_2(B) \;\Rightarrow\; 6 - 6p = 2 \;\Rightarrow\; p = \tfrac{2}{3}, \tag{1}$$

and player 1 is indifferent when

$$u_1(C) = 8q \;=\; 6q + 2(1-q) = u_1(D) \;\Rightarrow\; 8q = 2 + 4q \;\Rightarrow\; q = \tfrac{1}{2}. \tag{2}$$

The unique equilibrium of the subgame is $(p, q) = (\frac{2}{3}, \frac{1}{2})$, with values

$$u_1 = 8\left(\tfrac{1}{2}\right) = 4, \qquad u_2 = 2.$$

Step 2: subgame-perfect equilibrium. Replace x_1 by $(4, 2)$. At x_0, R gives $4 > 3$, so player 1 risks. Since the subgame has a unique equilibrium, the SPE is unique:

$$\sigma^{\text{SPE}}: \text{ player 1 plays } R \text{ and then } C \text{ w.p. } \tfrac{2}{3};\ \text{player 2 plays } A \text{ w.p. } \tfrac{1}{2}, \qquad \text{payoffs } (4, 2).$$

In the normal form this is player 1 mixing $\frac{2}{3}RC + \frac{1}{3}RD$ against $\frac{1}{2}A + \frac{1}{2}B$. There is *no* pure-strategy SPE.

Step 3: all Nash equilibria. Given q, player 1's payoffs are 3 (SC, SD), $8q$ (RC) and $2 + 4q$ (RD). Player 2's gain from A over B is 0 against SC, SD, -2 against RC and $+4$ against RD.

- $q \le \frac{1}{4}$: $3 \ge 2 + 4q$ and $3 > 8q$, so S is a best reply (strictly, for $q < \frac{1}{4}$). With player 1 on $\{SC, SD\}$ in any mix, player 2 is indifferent, so any such q is a best reply. *Nash equilibria.*
- $\frac{1}{4} < q < \frac{1}{2}$: RD is the unique best reply ($2 + 4q > 3$ and $2 + 4q > 8q$); then player 2 strictly prefers A, i.e. $q = 1$. Contradiction.
- $q = \frac{1}{2}$: RC and RD tie at $4 > 3$; player 2 is indifferent only if $-2\Pr(RC) + 4\Pr(RD) = 0$, i.e. $\frac{2}{3}RC + \frac{1}{3}RD$. *The SPE above.*
- $q > \frac{1}{2}$: RC is the unique best reply; then player 2 plays B, $q = 0$. Contradiction.

So the Nash equilibria are

$$\left\{ \text{(any mix of } SC, SD;\ q) : 0 \le q \le \tfrac{1}{4} \right\} \;\cup\; \left\{ \left(\tfrac{2}{3}RC + \tfrac{1}{3}RD;\ q = \tfrac{1}{2}\right) \right\},$$

and the *pure* Nash equilibria are (SC, B) and (SD, B), both with payoffs $(3, 3)$. None of the first family is subgame perfect: each asks player 2 to play A with probability at most $\frac{1}{4}$ in a subgame whose only equilibrium has $q = \frac{1}{2}$. The "threat" to play B deters R, but it is not credible once R has been played.

Interpretation. Playing safe is supported only by player 2's threat to play B in the risky game. Once the risky game starts, both must play its (mixed) equilibrium, which gives player 1 an expected 4, more than the safe 3, so he takes the risk. Player 2 is worse off ($2 < 3$): she would like to promise B, but cannot.

Verification. `check_B_L03.py` confirms that the subgame has no pure equilibrium, solves (1)–(2) with sympy, finds the pure Nash equilibria (SC, B), (SD, B) of the 4×2 matrix, and scans q over a grid of 241 points plus the critical values $\frac{1}{4}, \frac{3}{8}, \frac{1}{2}$. A Nash equilibrium exists exactly at $q \in [0, \frac{1}{4}]$ and $q = \frac{1}{2}$, and at $q = \frac{1}{2}$ the indifference forces $(\frac{2}{3}, \frac{1}{3})$ on (RC, RD). Tadelis prints no answer.

Answer. (a) Tree above; two proper subgames (the whole game and the simultaneous game after R). (b) Yes: player 2 can be drawn first in the stage game; a tree with root choices $\{S, RC, RD\}$ gives the same reduced normal form but no proper subgame other than the whole game. (c) Matrix above, $S_1 = \{SC, SD, RC, RD\}$. (d) Pure Nash: (SC, B), (SD, B); all Nash: player 1 on S (any mix of SC, SD) with $\Pr(A) \leq \frac{1}{4}$, plus $(\frac{2}{3}RC + \frac{1}{3}RD,\ \frac{1}{2}A + \frac{1}{2}B)$. Unique SPE: R, then C w.p. $\frac{2}{3}$; A w.p. $\frac{1}{2}$; payoffs $(4, 2)$. No pure SPE.

16 Tadelis Exercise 8.13

Exam · block L3

Source: Tadelis Exercise 8.13, page 173–174 (statement in the book and in Exercise 16 of the list; not reproduced here).

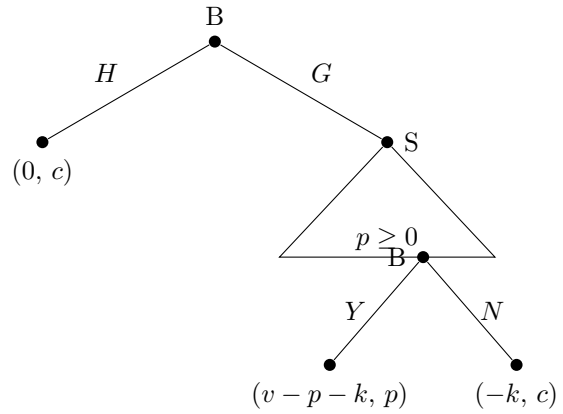
What is asked. A buyer pays a commuting cost k to reach a seller, who then names a take-it-or-leave-it price. (a) Extensive form and the buyer's reply at the acceptance node; (b) the SPE, its uniqueness and Pareto optimality; (c) Nash equilibria better for both; (d) whether the seller would pay ε to mail a postcard that commits him to a price.

Method. Backward induction with a continuum of actions (Tadelis §8.1 and §8.3.2, the Stackelberg logic, pp. 160–163); the ultimatum-game argument for ties (Tadelis Prop. 11.2, p. 223: the responder must accept at indifference, or the proposer has no best offer). Nash equilibria sustained by off-path threats (Tadelis Def. 7.8, p. 146). Commitment as a move added before the game (`Notas/L03/L03-06`, Cabral §7.2).

Solution. **Notation.** Buyer (B) and seller (S). B chooses G (go to the store) or H (stay home); then S names $p \geq 0$; then B accepts (Y) or walks out (N). Data: $c > 0$, $k > 0$, $v > c + k$. Payoffs (u_B, u_S):

$$H:\ (0,\ c); \qquad G, p, Y:\ (v - p - k,\ p); \qquad G, p, N:\ (-k,\ c).$$

(a) Extensive form and the buyer's reply.



The triangle stands for the continuum of prices; one typical price p is drawn.

At the node after price p, the commuting cost is already sunk; it appears in both payoffs. B accepts if

$$v - p - k \;\geq\; -k \iff p \leq v.$$

So the best reply is: accept every $p < v$, reject every $p > v$, and either action at $p = v$.

(b) SPE, uniqueness, efficiency.

Step 1: the acceptance rule at $p = v$. Suppose B rejected $p = v$. Then S would get p for every $p < v$ and $c < v$ otherwise, so his payoff approaches v as $p \uparrow v$ but never reaches it; S would have no best offer, so the profile cannot be an SPE. Hence in every SPE B accepts if and only if $p \leq v$.

Step 2: the seller. Given that rule, S's payoff is p for $p \leq v$ and $c < v$ for $p > v$; the unique maximizer is $p^* = v$.

Step 3: the buyer's commuting decision. Going yields

$$u_B(G) = v - p^* - k = v - (v) - k = -k \;<\; 0 = u_B(H),$$

so B stays home. Every step had a unique optimum, so the SPE is unique:

B: H, and accept iff $p \leq v$; S: offer $p = v$. Outcome: no trade, payoffs $(0,\ c)$.

Not Pareto optimal. Trade at any price $p \in (c,\ v - k)$ (nonempty since $v > c + k$) gives $u_B = v - p - k > 0$ and $u_S = p > c$: both strictly better. The lost surplus is $v - c - k > 0$. This is a hold-up: the commuting cost is sunk when the price is set, so S extracts all of v and B, anticipating it, never makes the trip.

(c) Better Nash equilibria. Yes. Fix $\hat{p} \in (c,\ v - k)$ and consider

B: G, accept iff $p \leq \hat{p}$; S: offer $\hat{p}$.

S's best reply: offers $p \leq \hat{p}$ are accepted and give $p \leq \hat{p}$; offers above are rejected and give $c < \hat{p}$. So $\hat{p}$ is optimal. *B's best reply:* given $\hat{p}$, going and accepting gives $v - \hat{p} - k > 0$, more than staying (0) or going and rejecting $(-k)$; B's rule at prices other than $\hat{p}$ is off the path and does not affect his payoff. Payoffs $(v - \hat{p} - k,\ \hat{p})$ beat $(0, c)$ for both. It is not subgame perfect: after an off-path offer $p \in (\hat{p}, v)$, rejecting gives $-k < v - p - k$, so the threat to walk out is not credible.

(d) The postcard. Add a first move: S either mails a postcard committing to $\hat{p}$ (at cost ε) or does not. Without the postcard the game is the one in (b): payoff c to S. With it, B buys at $\hat{p}$ if he goes and $\hat{p} \leq v$, so he goes iff $v - \hat{p} - k \geq 0$, i.e. $\hat{p} \leq v - k$. By the argument of Step 1, B goes at indifference. Then S's payoff from the postcard is

$$u_S(\hat{p}) = \begin{cases} \hat{p} - \varepsilon, & \hat{p} \leq v - k, \\ c - \varepsilon, & \hat{p} > v - k, \end{cases} \qquad \Rightarrow \qquad \hat{p}^* = v - k, \quad u_S = v - k - \varepsilon.$$

Compared with not mailing,

$$(v-k-\varepsilon)-c=(v-c-k)-\varepsilon>0 \quad \text{since } \varepsilon<v-c-k.$$

So in the SPE the seller mails the postcard with price $v-k$; B goes and buys; payoffs $(0,\ v-k-\varepsilon)$. Trade now happens; the total surplus is $v-c-k-\varepsilon$, all of it to the seller.

Interpretation. The commitment in (c) has to come from the buyer and is not credible; the commitment in (d) comes from the seller, who limits his own future options, and it is credible because the coupon binds him. "Junk mail" has value even when it contains no information: it is a commitment not to hold up a customer who has sunk a cost. The seller still captures the whole surplus; the coupon only removes the inefficiency, the same logic as the bond in the entry game (Cabral §7.2).

Verification. `check_B_L03.py` uses $c=2$, $v=10$, $k=1$, $\varepsilon=\frac{1}{2}$ and prices on a 0.25 grid: the seller's best offer is $p=10=v$ and the buyer stays home; under "reject at v" the best grid offer is $v-0.25$ (no maximum on a continuum); thresholds $\hat{p}\in\{2.5,5,8.5\}$ give Nash equilibria that beat $(0,c)$ for both; with the postcard, $\hat{p}^*=9=v-k$ and $8.5>2$; sympy confirms the gain $(v-c-k)-\varepsilon$. Tadelis prints no answer.

Answer. (a) Tree above; at the acceptance node the buyer accepts iff $p\le v$ (the sunk k does not matter). (b) Unique SPE: seller offers $p=v$, buyer accepts iff $p\le v$ and therefore stays home; payoffs $(0,c)$; not Pareto optimal, since trade at any $p\in(c,v-k)$ helps both. (c) Yes: buyer goes and accepts only $p\le\hat{p}$, seller offers $\hat{p}$, for any $\hat{p}\in(c,v-k)$; Nash but not subgame perfect. (d) Yes: the seller mails a coupon at $\hat{p}=v-k$, the buyer comes and buys, seller earns $v-k-\varepsilon>c$.

17 Muñoz-Garcia Ch. 4 Exercise 12

Exam · block L3

Source: Muñoz-Garcia Ch. 4 Exercise 12, page 185–186 (statement in the book and in Exercise 17 of the list; not reproduced here).

What is asked. With $p(Q)=10-Q$ and marginal cost 2: (a) the Stackelberg equilibrium with leader 1 and follower 2; (b) the equilibrium when a third firm moves after firm 2, and how firms 1 and 2's outputs change.

Method. Backward induction with continuous actions: solve the last mover's best reply, substitute it into the previous mover's profit, repeat (Tadelis §8.3.2, eqs. (8.1)–(8.2), pp. 160–161; Gibbons §2.1.B, pp. 61–64). The SPE lists each follower's whole reaction function, the outcome only the quantities (Gibbons pp. 125–126).

Solution. **Notation.** $q_i\ge 0$ is firm i's output, Q the total; price-cost margin $p-2=8-Q$; profit $\pi_i=(8-Q)\,q_i$.

(a) Two firms.

Step 1: follower. Firm 2 maximizes $\pi_2=(8-q_1-q_2)\,q_2$:

$$\frac{\partial\pi_2}{\partial q_2}=8-q_1-2q_2=0 \;\Rightarrow\; q_2(q_1)=4-\frac{q_1}{2} \quad (q_1\le 8). \tag{3}$$

Step 2: leader. Substituting (3) into π_1:

$$\pi_1 = \Big(8 - q_1 - (4 - \tfrac{q_1}{2})\Big)q_1 = \Big(4 - \frac{q_1}{2}\Big)q_1, \qquad \frac{d\pi_1}{dq_1} = 4 - q_1 = 0 \;\Rightarrow\; q_1^* = 4.$$

Step 3: outcome. From (3), $q_2^* = 4 - \frac{(4)}{2} = 2$; $Q = 6$, $p = 10 - 6 = 4$, and

$$\pi_1 = (4-2)(4) = 8, \qquad \pi_2 = (4-2)(2) = 4.$$

SPE: $(q_1^* = 4,\ q_2(q_1) = \max\{0, 4 - q_1/2\})$.

(b) Three firms, moving in the order 1, 2, 3.

Step 1: firm 3. It maximizes $(8 - q_1 - q_2 - q_3)\,q_3$:

$$8 - q_1 - q_2 - 2q_3 = 0 \;\Rightarrow\; q_3(q_1, q_2) = 4 - \frac{q_1 + q_2}{2}. \tag{4}$$

Step 2: firm 2. Substituting (4) into π_2:

$$\pi_2 = \Big(8 - q_1 - q_2 - (4 - \tfrac{q_1+q_2}{2})\Big)q_2 = \Big(4 - \frac{q_1 + q_2}{2}\Big)q_2,$$

$$\frac{\partial \pi_2}{\partial q_2} = 4 - \frac{q_1}{2} - q_2 = 0 \;\Rightarrow\; q_2(q_1) = 4 - \frac{q_1}{2}. \tag{5}$$

Substituting (5) into (4):

$$q_3(q_1) = 4 - \frac{q_1 + (4 - \frac{q_1}{2})}{2} = 2 - \frac{q_1}{4}.$$

Step 3: firm 1. Substituting both:

$$\pi_1 = \Big(8 - q_1 - (4 - \tfrac{q_1}{2}) - (2 - \tfrac{q_1}{4})\Big)q_1 = \Big(2 - \frac{q_1}{4}\Big)q_1, \qquad \frac{d\pi_1}{dq_1} = 2 - \frac{q_1}{2} = 0 \;\Rightarrow\; q_1^* = 4.$$

Step 4: outcome. $q_2^* = 4 - \frac{(4)}{2} = 2$, $q_3^* = 2 - \frac{(4)}{4} = 1$; $Q = 7$, $p = 3$, margin 1:

$$\pi_1 = 4, \qquad \pi_2 = 2, \qquad \pi_3 = 1.$$

Comparison. Firms 1 and 2 produce exactly what they produced in (a), 4 and 2; firm 3 adds 1. Total output rises from 6 to 7, the price falls from 4 to 3, and the profits of firms 1 and 2 halve (8 to 4, and 4 to 2).

Interpretation. Why do the outputs not move? Firm 3 absorbs half of any change in $q_1 + q_2$ (slope $-\frac{1}{2}$ in (4)). After substituting it, firm 2's residual margin is $4 - \frac{q_1+q_2}{2}$, half of the margin it faced in (a): the same problem scaled by $\frac{1}{2}$, so the same maximizer. The same scaling repeats upstream, which gives the pattern $q_k = (a-c)/2^k$ for the k-th mover with demand $a - Q$ and marginal cost c (here $a - c = 8$: $4, 2, 1$). The leader always produces the monopoly output $(a-c)/2 = 4$. Entry at the end of the chain does not change earlier outputs, but it lowers their profits through the price.

Verification. `check_B_L03.py` redoes every first-order condition with sympy (reaction functions, $q^* = (4, 2)$ and $(4, 2, 1)$, prices 4 and 3, profits $(8, 4)$ and $(4, 2, 1)$, concavity of the leader's problem) and repeats both games numerically with the leader on a 0.01 grid. *Comparison with Muñoz-Garcia's answer* (pp. 186–189): same quantities, $(4, 2)$ and $(4, 2, 1)$, and the same conclusion that firms 1 and 2 do not change output. The book reports no prices or profits. Minor slips in its text: the interior condition of firm 2's first-order condition is written "if $q_1 > 0$" where $q_2 > 0$ is meant, both in (a) and in (b).

Answer. (a) $q_1 = 4$, $q_2 = 4 - q_1/2 = 2$; $p = 4$; $\pi = (8, 4)$. (b) $q_1 = 4$, $q_2 = 4 - q_1/2 = 2$, $q_3 = 4 - (q_1 + q_2)/2 = 1$; $p = 3$; $\pi = (4, 2, 1)$. Firms 1 and 2's outputs are unchanged by firm 3's entry; their profits halve because the price falls.

18 Gibbons Problem 2.1

Advanced · block L3

Source: Gibbons Problem 2.1, page 130 (statement in the book and in Exercise 18 of the list; not reproduced here).

What is asked. Prove Becker's Rotten Kid theorem: when a child's action A produces incomes $I_C(A)$, $I_P(A)$ and an altruistic parent then chooses a bequest B, the backwards-induction outcome has the child choosing the A that maximizes $I_C(A) + I_P(A)$.

Method. Backward induction in a two-stage game with continuous actions (Gibbons §2.1.A, pp. 57–61; Tadelis §8.3.2): solve the parent's problem for every pair of incomes, then the child's problem given the parent's reply. The key step is that the parent's reply depends on the incomes only through their sum, and raises the child's consumption when the sum rises.

Solution. Notation and assumptions. $A \geq 0$ the child's action; I_C, I_P strictly concave with maximizers $A_C, A_P > 0$; $B \in \mathbb{R}$ the bequest; U, V increasing and strictly concave; $k > 0$. Write $Y = I_C + I_P$ (family income) and $x = I_C + B$ (the child's consumption), so the parent consumes $I_P - B = Y - x$. As is standard for this problem, U and V are differentiable and the parent's problem has a solution.

Step 1: the parent's problem depends only on Y. Given (I_C, I_P) the parent solves

$$\max_{B \in \mathbb{R}} \; V(I_P - B) + k\,U(I_C + B).$$

Because B is unrestricted in sign, choosing B is the same as choosing $x = I_C + B \in \mathbb{R}$, and the problem becomes

$$\max_{x \in \mathbb{R}} \; W(x; Y) = V(Y - x) + k\,U(x). \tag{6}$$

W is strictly concave in x (a sum of strictly concave functions), so the maximizer $x^*(Y)$ is unique and satisfies the first-order condition

$$V'(Y - x^*) = k\,U'(x^*). \tag{7}$$

The bequest is $B^*(I_C, I_P) = x^*(I_C + I_P) - I_C$: it depends on both incomes, but the child's consumption depends only on their sum.

Step 2: x^* is strictly increasing in Y. Take $Y' > Y$ and suppose $x' = x^*(Y') \leq x = x^*(Y)$. Then $Y' - x' > Y - x$. Strict concavity makes V' strictly decreasing and U' decreasing, so

$$V'(Y' - x') < V'(Y - x) \qquad \text{and} \qquad k\,U'(x') \geq k\,U'(x).$$

Applying (7) at Y and at Y':

$$V'(Y' - x') = k\,U'(x') \;\geq\; k\,U'(x) = V'(Y - x) \;>\; V'(Y' - x'),$$

a contradiction. Hence $x^*(Y') > x^*(Y)$. (With U, V twice differentiable, $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$: the parent passes part, not all, of an income gain to the child.)

Step 3: the child's problem. Anticipating B^*, the child chooses $A \geq 0$ to maximize

$$U\big(I_C(A) + B^*(I_C(A), I_P(A))\big) = U\big(x^*(I_C(A) + I_P(A))\big) = U\big(x^*(Y(A))\big).$$

$U \circ x^*$ is strictly increasing (Steps 1–2 and U increasing), so A maximizes the child's payoff if and only if it maximizes $Y(A) = I_C(A) + I_P(A)$.

Step 4: the maximizer exists and is unique. Y is strictly concave. For $A < \min\{A_C, A_P\}$ both incomes are increasing, and for $A > \max\{A_C, A_P\}$ both are decreasing, so Y attains its maximum on the compact interval $[\min\{A_C, A_P\}, \max\{A_C, A_P\}] \subset (0, \infty)$, at a unique point A^*. The backwards-induction outcome is

$$(A^*,\ B^*(I_C(A^*), I_P(A^*))), \qquad A^* = \arg\max_{A \geq 0}\ [\, I_C(A) + I_P(A)\,]. \qquad \square$$

Interpretation. The child is selfish but faces a parent who "taxes" any loss of family income: if the child raised I_C at a larger cost to I_P, the parent, who cares about the split, would cut the bequest by more than the child gained. Altruism on one side, plus the parent's last move, aligns the first mover's incentives with joint surplus. The theorem needs the parent to move last with an unrestricted transfer: if $B \geq 0$ were imposed and binding, or if the child moved after the bequest, Step 1 would fail. Note that the child does *not* pick A_C; it is between A_C and A_P.

Verification. `check_B_L03.py` checks the sign of dx^*/dY for several values of (V'', U'', k), and solves an example numerically: $U = V = -e^{-x}$, $k = 2$, $I_C = 4A - A^2$ ($A_C = 2$), $I_P = 10A - A^2$ ($A_P = 5$). For each A on a 0.05 grid it finds the parent's best bequest on a fine grid; the child's best action is $A = 3.5$, the maximizer of $I_C + I_P = 14A - 2A^2$, not $A_C = 2$. The child's consumption matches the closed form $x^* = (Y + \ln k)/2$, which is increasing in Y. Gibbons prints no answer.

Answer. The parent's best bequest gives the child consumption $x^*(I_C + I_P)$ that solves $V'(Y - x) = kU'(x)$; it depends only on family income Y and is strictly increasing in it. So the child's payoff $U(x^*(Y(A)))$ is a strictly increasing function of $Y(A)$, and the child chooses $A^* = \arg\max_A [I_C(A) + I_P(A)]$, which exists and is unique by strict concavity.

Lecture 4 — Multistage games and bargaining

19 Tadelis Exercise 9.1

Foundations · block L4

Source: Tadelis Exercise 9.1, page 186–187 (statement in the book and in Exercise 19 of the list; not reproduced here).

What is asked. The 3×3 game is played twice with the first-period outcome observed. (a) All pure-strategy SPE with $\delta = 1$, with history-contingent strategies; (b) for each, the smallest δ that supports it.

Method. Multistage games (Tadelis §9.1–9.4, pp. 175–184): period-2 play must be a stage Nash equilibrium after every history (Prop. 9.2, p. 181), and a non-equilibrium first-period profile can be supported only with a carrot (a good stage equilibrium on the path) and a stick (a bad one after a deviation), when $\delta \geq \delta^* = g/(c - s)$ (`Notas/L04/L04-02`). Deviations are checked one stage at a time (one-stage deviation principle, Thm. 9.1, p. 185).

Solution. Notation. $v_i = u_i(a^1) + \delta\, u_i(a^2)$, where a^t is the period-t action profile. A pure strategy for player i is a period-1 action and a map from the nine possible period-1 outcomes to a period-2 action. $g_i(a)$ is player i's best one-shot gain from deviating from a:

$$g_1(a) = \max_r u_1(r, a_2) - u_1(a), \qquad g_2(a) = \max_c u_2(a_1, c) - u_2(a).$$

Step 1: stage equilibria. Best replies of player 1: to L, B (13); to C, M (5); to R, B (1). Of player 2: to T, R (13); to M, C (5); to B, R (1). The mutual best replies are

$$(M, C) \text{ with } (5, 5) \quad \text{(the carrot)}, \qquad (B, R) \text{ with } (1, 1) \quad \text{(the stick)}.$$

Each player gets 5 in the carrot and 1 in the stick, so the largest punishment is $c - s = 5 - 1 = 4$.

Step 2: which first-period profiles can be supported. Period-2 play is (M, C) or (B, R) after every history. A first-period profile a is supportable if and only if each player's gain is at most what the threat can take away:

$$g_i(a) \;\le\; \delta\,(5 - 1) = 4\delta, \qquad i = 1, 2, \tag{8}$$

using (M, C) after a and (B, R) after any deviation. No continuation can do better, since 5 is the most and 1 the least a player can get in period 2. For a stage equilibrium a, $g_i(a) = 0$ and any continuation works. The gains are:

a	$u(a)$	$g_1(a)$	$g_2(a)$	supportable ($\delta = 1$)?	$\delta^* = \max_i g_i/4$
(T, L)	$(10, 10)$	$13 - 10 = 3$	$13 - 10 = 3$	yes	$3/4$
(T, C)	$(2, 12)$	$5 - 2 = 3$	$13 - 12 = 1$	yes	$3/4$
(T, R)	$(0, 13)$	$1 - 0 = 1$	0	yes	$1/4$
(M, L)	$(12, 2)$	$13 - 12 = 1$	$5 - 2 = 3$	yes	$3/4$
(M, C)	$(5, 5)$	0	0	yes	0
(M, R)	$(0, 0)$	$1 - 0 = 1$	$5 - 0 = 5$	no	–
(B, L)	$(13, 0)$	0	$1 - 0 = 1$	yes	$1/4$
(B, C)	$(0, 0)$	$5 - 0 = 5$	$1 - 0 = 1$	no	–
(B, R)	$(1, 1)$	0	0	yes	0

(a) The pure SPE with $\delta = 1$. The SPE *paths* are nine:

- the four sequences of stage equilibria, played regardless of history (Prop. 9.1, p. 180): (M, C) then (M, C); (M, C) then (B, R); (B, R) then (M, C); (B, R) then (B, R);
- the five carrot-and-stick paths: a then (M, C), for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$.

For a carrot-and-stick path the strategies are, for $\hat{a} = (\hat{a}_1, \hat{a}_2)$ one of those five:

$$s_1 : \text{ play } \hat{a}_1; \text{ in period 2 play } M \text{ if } \hat{a} \text{ was played, } B \text{ otherwise;}$$
$$s_2 : \text{ play } \hat{a}_2; \text{ in period 2 play } C \text{ if } \hat{a} \text{ was played, } R \text{ otherwise.}$$

For example, for (T, L): on the path each gets $10 + 5 = 15$; player 1's best deviation (B) gives $13 + 1 = 14$, player 2's (R) gives $13 + 1 = 14$. In period 2 both play a stage equilibrium after every history, so no deviation pays there. (M, R) and (B, C) cannot be supported: a gain of 5 exceeds the largest punishment 4.

These strategies are not the only ones. Any assignment of (M, C) or (B, R) to the nine period-1 outcomes that satisfies (8) for the deviations that would be profitable also works (e.g. for (T, R) only player 1's deviation to B must be punished). Counting every such map, `check_B_L04.py` finds 992 pure SPE strategy profiles at $\delta = 1$, all with one of the nine paths above.

(b) The smallest discount factor. From (8), $\delta^*(a) = \max_i g_i(a)/4$ (last column): $\frac{3}{4}$ for (T, L), (T, C), (M, L); $\frac{1}{4}$ for (T, R), (B, L); and any $\delta \ge 0$ for the four sequences of stage equilibria. For (T, L): $10 + 5\delta \ge 13 + \delta \iff \delta \ge \frac{3}{4}$.

Interpretation. Two stage equilibria with different payoffs give the second period a carrot and a stick, and that lets the players reach (T, L), which pays $(10, 10)$, twice the $(5, 5)$ of the good stage equilibrium, in the first period. Profiles with a gain above 4 cannot be enforced by a single future period, however patient the players are.

Verification. `check_B_L04.py` finds the two stage equilibria, enumerates all 9×2^9 combinations of first-period profile and continuation map at $\delta = 1$ (992 SPE, 9 paths; (M, R) and (B, C) excluded), and checks every threshold: each path is an SPE at δ^*, and no continuation map at all supports it at $\delta^* - 0.001$. Tadelis prints no answer.

Answer. Stage equilibria: (M, C) $(5, 5)$ and (B, R) $(1, 1)$. (a) Pure SPE paths with $\delta = 1$: the four sequences of stage equilibria, and a then (M, C) for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$, supported by "play (M, C) after a, (B, R) after any other outcome". (M, R) and (B, C) cannot be supported. (b) $\delta^* = \frac{3}{4}$ for (T, L), (T, C), (M, L); $\frac{1}{4}$ for (T, R), (B, L); 0 for the sequences of stage equilibria.

20 Gibbons Problem 2.6

Foundations · block L4

Source: Gibbons Problem 2.6, page 133 (statement in the book and in Exercise 20 of the list; not reproduced here).

What is asked. Firm 1 chooses q_1; firms 2 and 3 observe it and choose q_2, q_3 simultaneously; demand $P = a - Q$, marginal cost c. Find the subgame-perfect outcome.

Method. Two-stage game of complete but imperfect information (Gibbons §2.2.A, pp. 71–73): for each q_1 solve the second-stage simultaneous game for its Nash equilibrium, then let firm 1 optimize against it. The second stage is a Cournot duopoly on the residual demand (Gibbons §1.2.A); the first stage is Stackelberg logic (Gibbons §2.1.B; Tadelis §8.3.2).

Solution. **Notation.** $a > c \geq 0$; $q_i \geq 0$; profit $\pi_i = (a - Q - c)\, q_i$ with $Q = q_1 + q_2 + q_3$.

Step 1: second stage, given $q_1 < a - c$. Firm 2's and firm 3's first-order conditions:

$$\frac{\partial \pi_2}{\partial q_2} = a - c - q_1 - 2q_2 - q_3 = 0, \tag{9}$$

$$\frac{\partial \pi_3}{\partial q_3} = a - c - q_1 - q_2 - 2q_3 = 0. \tag{10}$$

Subtracting (10) from (9) gives $q_3 - q_2 = 0$, so $q_2 = q_3$. Substituting $q_3 = q_2$ into (9):

$$a - c - q_1 - 2q_2 - (q_2) = 0 \;\Rightarrow\; q_2(q_1) = q_3(q_1) = \frac{a - c - q_1}{3}.$$

This is the Cournot duopoly with demand intercept $a - q_1$; its equilibrium is unique. If $q_1 \geq a - c$, both followers produce 0.

Step 2: first stage. Substituting the followers' outputs, for $q_1 < a - c$,

$$\pi_1 = \Big(a - q_1 - 2 \cdot \frac{a - c - q_1}{3} - c\Big) q_1 = \frac{a - c - q_1}{3}\, q_1,$$

a strictly concave function with

$$\frac{d\pi_1}{dq_1} = \frac{a - c - 2q_1}{3} = 0 \;\Rightarrow\; q_1^* = \frac{a - c}{2}.$$

For $q_1 \geq a - c$ the price is at most c, so $\pi_1 \leq 0$; the interior choice is the global one.

Step 3: outcome.

$$q_2^* = q_3^* = \frac{a - c - \frac{a-c}{2}}{3} = \frac{a - c}{6}, \qquad Q^* = \frac{a - c}{2} + 2 \cdot \frac{a - c}{6} = \frac{5(a - c)}{6},$$

$$P^* = a - \frac{5(a-c)}{6} = \frac{a+5c}{6},$$

$$\pi_1^* = \frac{a-c}{6}\cdot\frac{a-c}{2} = \frac{(a-c)^2}{12}, \qquad \pi_2^* = \pi_3^* = \frac{a-c}{6}\cdot\frac{a-c}{6} = \frac{(a-c)^2}{36}.$$

The subgame-perfect outcome is $(q_1^*, q_2^*, q_3^*) = (\frac{a-c}{2}, \frac{a-c}{6}, \frac{a-c}{6})$. The SPE itself lists the followers' strategies $q_2(q_1) = q_3(q_1) = \max\{0, (a-c-q_1)/3\}$.

Interpretation. The leader again chooses the monopoly output $(a-c)/2$, as in the two-firm Stackelberg game (Gibbons p. 63) and in item 17: with linear demand, the followers' joint reply has slope $-\frac{2}{3}$, and the leader's residual margin $(a-c-q_1)/3$ is a scaled copy of the monopoly margin. Compared with simultaneous three-firm Cournot (each $(a-c)/4$, $Q = \frac{3}{4}(a-c)$), the leader produces more and earns more ($\frac{(a-c)^2}{12} > \frac{(a-c)^2}{16}$), the followers less, and total output is higher.

Verification. `check_B_L04.py` solves (9)–(10) and firm 1's problem with sympy and checks Q^*, P^* and both profits; it also solves the case $a = 13$, $c = 1$ with the leader on a grid of step 0.05: $q_1 = 6$, $q_2 = q_3 = 2$. Gibbons prints no answer.

Answer. $q_1^* = \frac{a-c}{2}$, $q_2^* = q_3^* = \frac{a-c}{6}$ (followers' rule $q_j(q_1) = \max\{0, \frac{a-c-q_1}{3}\}$); $P^* = \frac{a+5c}{6}$; profits $\frac{(a-c)^2}{12}$ for firm 1 and $\frac{(a-c)^2}{36}$ for each follower.

21 Tadelis Exercise 9.4

Exam · block L4

Source: Tadelis Exercise 9.4, page 188 (statement in the book and in Exercise 21 of the list; not reproduced here).

What is asked. A Cournot stage ($p = 100 - q$, zero cost) is followed by an "access" game with equilibria (O, o) and (N, n). (a) The Cournot equilibrium, the access game's pure equilibria and all pure SPE when $\delta = 0$; (b) an SPE with monopoly output split equally when $\delta = 1$; (c) the lowest δ for it; (d) whether it survives at $\delta = 0.4$, and if not the best symmetric profits.

Method. Multistage games with a carrot and a stick in the last stage (Tadelis §9.4, pp. 180–184; threshold $\delta^* = g/(c-s)$, `Notas/L04/L04-02`); Cournot best replies (Tadelis §5.2.3, pp. 87–88). Payoffs $v_i = \pi_i(q_1, q_2) + \delta\, u_i(\text{access})$.

Solution. (a) Stage games and $\delta = 0$.

Cournot. Firm i maximizes $(100 - q_i - q_j)\, q_i$:

$$100 - 2q_i - q_j = 0 \;\Rightarrow\; q_i = BR_i(q_j) = \frac{100 - q_j}{2}. \tag{11}$$

By symmetry $q_i = q_j = q$, so $q = \frac{100-q}{2}$, i.e. $q^C = \frac{100}{3}$, price $\frac{100}{3}$, and $\pi^C = (\frac{100}{3})^2 = \frac{10000}{9} \approx 1111.1$ each.

Access game. Payoffs (player 1 rows):

	o	n
O	2500, 2500	−10, 0
N	0, −10	0, 0

O is the best reply to o ($2500 > 0$) and N the best reply to n ($0 > -10$); symmetrically for player 2. Pure equilibria: (O, o) with $(2500, 2500)$ and (N, n) with $(0, 0)$.

SPE with $\delta = 0$. Stage-2 payoffs carry weight 0, so continuation play cannot reward or punish anything, and stage 1 must be the Cournot equilibrium. The pure SPE are

$$s_i: \; q_i = \tfrac{100}{3}; \text{ in stage 2, after each } (q_1, q_2), \text{ play the action of } e(q_1, q_2),$$

where e is *any* map from $\mathbb{R}^2_+$ to $\{(O, o), (N, n)\}$ (e.g. always (O, o)). There is a continuum of them, all with first-stage outcome $(\frac{100}{3}, \frac{100}{3})$. Strictly, with $\delta = 0$ the stage-2 actions do not affect any payoff, so every stage-2 profile is an equilibrium of those subgames; the answer above uses the standard reading that stage-2 actions are judged by stage-2 payoffs (the limit $\delta \to 0^+$).

(b) Monopoly split with $\delta = 1$. Joint profit $(100 - Q)Q$ is maximized at $Q = 50$; each firm produces 25 and earns

$$\pi^M = 25\,(100 - 50) = 1250.$$

Strategies (for both firms):

$$s_i: \; q_i = 25; \text{ in stage 2, open } (O/o) \text{ if } (q_1, q_2) = (25, 25), \text{ keep closed } (N/n) \text{ otherwise.}$$

Stage 2. After every history the prescribed profile is an equilibrium of the access game. *Stage 1.* The best deviation against $q_j = 25$ is, by (11), $q_i = \frac{100-25}{2} = 37.5$, with price $100 - 62.5 = 37.5$ and

$$\pi^D = 37.5 \times 37.5 = 1406.25.$$

Staying gives $1250 + \delta\,(2500)$; deviating gives $1406.25 + \delta\,(0)$. With $\delta = 1$: $3750 \geq 1406.25$. It is an SPE.

(c) Lowest δ.

$$1250 + 2500\,\delta \;\geq\; 1406.25 \iff \delta \;\geq\; \delta^* = \frac{156.25}{2500} = \frac{1}{16} = 0.0625. \tag{12}$$

In the carrot-and-stick form: $g = 156.25$, $c - s = 2500 - 0$.

(d) $\delta = 0.4$. Since $0.4 > \frac{1}{16}$, condition (12) holds: $1250 + 0.4\,(2500) = 2250 \geq 1406.25$. The monopoly split *is* supported, with the same strategies as in (b), and each firm earns 2250 in discounted total. The "if not" branch does not arise with these numbers.

For completeness, when $\delta < \frac{1}{16}$: with symmetric output q and the same threat, the deviation gain is

$$g(q) = \Big(\frac{100 - q}{2}\Big)^2 - q\,(100 - 2q) = \frac{9}{4}\Big(\frac{100}{3} - q\Big)^2,$$

and $g(q) \leq 2500\,\delta$ holds iff $q \geq \frac{100}{3}(1 - \sqrt{\delta})$. The best symmetric equilibrium uses $q = \max\{25, \; \frac{100}{3}(1 - \sqrt{\delta})\}$; e.g. at $\delta = 0.04$, $q = \frac{80}{3}$ and stage-1 profit $\frac{11200}{9} \approx 1244.4$ each.

Interpretation. The second stage is worth far more (2500) than the one-shot gain from cheating on the cartel (156.25), so even very impatient firms ($\delta \geq 0.0625$) can collude in stage 1. The threat is credible because (N, n) is itself an equilibrium: once someone cheats, neither firm wants to be the only one to open. A future coordination game with a good and a bad equilibrium is an effective enforcement device, which is the multistage logic of Tadelis §9.4.

Verification. `check_B_L04.py` recomputes the Cournot equilibrium $(\frac{100}{3}, \frac{10000}{9})$, the two pure equilibria of the access game, $\pi^M = 1250$, $\pi^D = 1406.25$, $\delta^* = \frac{1}{16}$, the total 2250 at $\delta = 0.4$, the identity for $g(q)$, the bound $q \geq \frac{100}{3}(1 - \sqrt{\delta})$ (equal to 25 at $\delta = \frac{1}{16}$), and the $\delta = 0.04$ example. Tadelis prints no answer.

Answer. (a) Cournot $q_i = \frac{100}{3}$, $\pi_i = \frac{10000}{9}$; access game pure equilibria (O, o) and (N, n); with $\delta = 0$ every pure SPE has $q_i = \frac{100}{3}$ followed by (O, o) or (N, n) after each history (any assignment). (b) $q_i = 25$; open after $(25, 25)$, closed otherwise. (c) $\delta^* = \frac{1}{16}$. (d) Yes, $0.4 > \frac{1}{16}$: monopoly split supported, discounted payoff 2250 each.

22 Tadelis Exercise 11.2

Exam · block L4

Source: Tadelis Exercise 11.2, page 236 (statement in the book and in Exercise 22 of the list; not reproduced here).

What is asked. Player 2 invests L or H (cost $c_L < c_H$, pie $v_L < v_H$, $v_H - c_H > v_L - c_L$), then player 1 makes an ultimatum offer. (a) The unique SPE and whether it is Pareto optimal; (b) a Nash equilibrium better for both players.

Method. The ultimatum game's unique SPE (Tadelis Prop. 11.2, p. 223: the responder accepts every offer, the proposer offers nothing) and its many Nash equilibria (Prop. 11.1, p. 223), placed after an investment stage; backward induction over the two subgames (Tadelis §8.2). This is the hold-up problem (`Notas/L04/L04-05`).

Solution. Notation. After investment $j \in \{L, H\}$ the pie is v_j. Player 1 offers player 2 an amount $y \in [0, v_j]$ and keeps $v_j - y$ (Tadelis's shares $(x, 1-x)$ scaled by v_j). If 2 accepts, payoffs are $(v_j - y,\ y - c_j)$; if 2 rejects, $(0,\ -c_j)$. The investment cost is sunk when 2 responds.

(a) The SPE.

Step 1: each ultimatum subgame. After j, accepting y gives 2 $y - c_j$ and rejecting gives $-c_j$, so 2 accepts every $y > 0$ and is indifferent at $y = 0$. If 2 rejected $y = 0$, player 1 would want the smallest positive offer, which does not exist; so in every SPE 2 accepts every $y \geq 0$, and 1 offers $y = 0$ (Prop. 11.2). Payoffs in the subgame: $(v_j,\ -c_j)$.

Step 2: investment. Player 2 compares

$$u_2(L) = -c_L \ > \ -c_H = u_2(H),$$

so she invests low. The unique SPE is

$$2\text{: } L, \text{ accept every offer after } L \text{ and after } H; \qquad 1\text{: offer } y = 0 \text{ after } L \text{ and after } H,$$

with outcome L, $y = 0$, accepted; payoffs $(v_L,\ -c_L)$.

Not Pareto optimal. Take H followed by an accepted offer y. Both gain if

$$v_H - y > v_L \quad \text{and} \quad y - c_H > -c_L \iff c_H - c_L \ < \ y \ < \ v_H - v_L.$$

This interval is nonempty exactly because $v_H - c_H > v_L - c_L$. So the SPE outcome is Pareto dominated: the efficient investment is H, but its return is fully appropriated by player 1.

(b) A better Nash equilibrium. Fix $y^* \in (c_H - c_L,\ v_H - v_L)$:

$$2: \ H; \text{ after } H \text{ accept iff } y \geq y^*; \text{ after } L \text{ accept every offer;}$$
$$1: \text{ after } H \text{ offer } y^*; \text{ after } L \text{ offer } 0.$$

Player 1. After H, offers $y \geq y^*$ give $v_H - y$, maximized at y^* with $v_H - y^* > v_L > 0$; lower offers are rejected and give 0. So y^* is a best reply. (After L, which is off the path, offering 0 is

also optimal.) *Player 2.* H yields $y^* - c_H$; switching to L yields $0 - c_L$ (1 offers 0 after L). H is better since $y^* > c_H - c_L$. Accepting y^* beats rejecting ($y^* - c_H > -c_H$). Changing her rule at other offers changes nothing, because those offers are not made. Payoffs $(v_H - y^*,\ y^* - c_H)$ exceed $(v_L, -c_L)$ for both.

It is a Nash equilibrium but not an SPE: after H, rejecting an offer $y \in (0, y^*)$ gives $-c_H < y - c_H$, so the threat that holds player 1 to y^* is not credible.

Interpretation. Once the investment is sunk, the ultimatum stage gives player 1 the whole pie, so player 2 bears the full cost of investing and gets none of its return: she underinvests. Nash equilibrium can promise her a return through a threat to reject, but the threat is not credible. Credible fixes change the game: contracting on y before the investment, giving player 2 bargaining power ex post (alternating offers), or putting the investment and the pie under one owner (integration).

Verification. `check_B_L04.py` uses $v_L = 4$, $v_H = 10$, $c_L = 1$, $c_H = 3$: in the SPE player 2 invests low (payoffs $(4, -1)$); offers $y \in \{2.5, 4, 5.5\} \subset (2, 6)$ make both better off after H; and for those thresholds, player 1's best offer on a 0.25 grid is exactly y^* while player 2 prefers H. Tadelis prints no answer.

Answer. (a) Unique SPE: player 2 invests L and accepts every offer; player 1 offers 0 after either investment; payoffs $(v_L, -c_L)$. Not Pareto optimal: H with an accepted offer $y \in (c_H - c_L,\ v_H - v_L)$ makes both better off. (b) Yes: 2 plays H and accepts after H only offers $y \geq y^*$, 1 offers y^* after H and 0 after L, for any $y^* \in (c_H - c_L,\ v_H - v_L)$; payoffs $(v_H - y^*, y^* - c_H)$. It relies on a non-credible threat to reject.

23 Tadelis Exercise 11.5

Exam · block L4

Source: Tadelis Exercise 11.5, page 236–237 (statement in the book and in Exercise 23 of the list; not reproduced here).

What is asked. Three-period alternating-offer bargaining over a pie of 1, player 1 proposing first, with discount factors $\delta_1 \neq \delta_2$: (a) the SPE outcome; (b) player 1's advantage when $\delta_1 = \delta_2$; (c) when player 2 has the advantage, and why.

Method. Backward induction in finite alternating-offer bargaining (Tadelis §11.2, pp. 224–227; Gibbons §2.1.D, pp. 68–70): in the last period the proposer takes everything; in each earlier period the proposer offers the responder exactly the discounted value of what the responder would get next period, and the responder accepts at indifference (Prop. 11.2's argument, p. 223).

Solution. Notation. x_t = player 1's share of the pie when agreement is reached in period $t \in \{1, 2, 3\}$; player i's payoff from agreement $(x, 1-x)$ in period t is δ_i^{t-1} times his share. Players 1, 2, 1 propose in periods 1, 2, 3; no agreement in period 3 gives $(0, 0)$.

Step 1: period 3. Player 1 offers $x_3 = 1$ and player 2 accepts (she gets 0 either way). Discounted payoffs: $(\delta_1^2,\ 0)$.

Step 2: period 2. Player 2 must give player 1 at least what he gets by rejecting:

$$\delta_1\, x_2 \ \geq\ \delta_1^2\, x_3 = \delta_1^2 \ \Rightarrow\ x_2 = \delta_1. \tag{13}$$

Player 2 keeps $1 - x_2 = 1 - \delta_1$, worth $\delta_2(1 - \delta_1)$ in period-1 terms. (She prefers this to making an offer player 1 rejects, which gives her 0.)

Step 3: period 1. Player 1 must give player 2 at least her period-2 value:

$$1 - x_1 \;\geq\; \delta_2\,(1 - x_2) = \delta_2\,(1 - (\delta_1)),$$

so, keeping as much as possible,

$$x_1 = 1 - \delta_2(1 - \delta_1) = 1 - \delta_2 + \delta_1\delta_2. \tag{14}$$

(a) The SPE outcome is immediate agreement on

$$(v_1, v_2) = \big(1 - \delta_2(1 - \delta_1),\quad \delta_2(1 - \delta_1)\big).$$

(Strategies: player 1 offers x_1 in period 1 and 1 in period 3, and accepts in period 2 iff offered at least δ_1; player 2 accepts in period 1 iff offered at least $\delta_2(1 - \delta_1)$, offers $x_2 = \delta_1$ in period 2, accepts anything in period 3.)

(b) Equal discount factors. Setting $\delta_1 = \delta_2 = \delta$ in (14):

$$v_1 = 1 - \delta + \delta^2, \qquad v_2 = \delta - \delta^2,$$

$$v_1 - v_2 = 1 - 2\delta + 2\delta^2 = (1 - \delta)^2 + \delta^2 \;>\; 0.$$

So player 1 gets strictly more than half for every δ: he proposes first *and* last.

(c) When player 2 is ahead. Player 2 gets more than half iff

$$\delta_2\,(1 - \delta_1) > \tfrac{1}{2} \iff \delta_1 < 1 - \frac{1}{2\delta_2},$$

which needs $\delta_2 > \frac{1}{2}$ (a patient player 2) and δ_1 low enough (an impatient player 1). Example: $\delta_1 = 0.2$, $\delta_2 = 0.9$ gives $v_2 = 0.9 \times 0.8 = 0.72$. *Why.* Player 1's power comes from the last period, but an impatient player 1 values it at only δ_1 from period 2's viewpoint, so player 2 keeps $1 - \delta_1$ when she proposes. A patient player 2 values that period-2 position almost fully (δ_2), so in period 1 player 1 must concede $\delta_2(1 - \delta_1)$. Patience is bargaining power: v_1 rises with δ_1 ($\partial v_1/\partial\delta_1 = \delta_2 > 0$) and falls with δ_2 ($\partial v_1/\partial\delta_2 = -(1 - \delta_1) < 0$), as in the infinite-horizon formula $\frac{1-\delta_2}{1-\delta_1\delta_2}$ (Tadelis §11.3; `Notas/L04/L04-07`).

Interpretation. First-mover and last-mover advantages favour player 1, so only a large patience gap in player 2's favour can overturn them. With equal patience player 1's edge $(1 - \delta)^2 + \delta^2$ is smallest at $\delta = \frac{1}{2}$ (edge $\frac{1}{2}$, $v_1 = \frac{3}{4}$). As $\delta \to 0$ the first-mover advantage takes over and $v_1 \to 1$; as $\delta \to 1$ the last-mover advantage takes over and again $v_1 \to 1$, because waiting for period 3 costs player 1 nothing.

Verification. `check_B_L04.py` redoes the recursion symbolically ($x_1 = 1 - \delta_2 + \delta_1\delta_2$; $v_1 - v_2 = (1 - \delta)^2 + \delta^2$), tests the condition in (c) on four (δ_1, δ_2) pairs, and repeats the backward induction numerically with offers on a grid of step $\frac{1}{1000}$ for three pairs, e.g. $(\delta_1, \delta_2) = (0.9, 0.7)$ gives $x_1 = 0.93$, the value in `Notas/L04/L04-10-formulas.md`. Tadelis prints no answer.

Answer. (a) Immediate agreement: player 1 gets $1 - \delta_2(1 - \delta_1)$, player 2 gets $\delta_2(1 - \delta_1)$. (b) With $\delta_1 = \delta_2 = \delta$: $v_1 - v_2 = (1 - \delta)^2 + \delta^2 > 0$. (c) Player 2 is ahead iff $\delta_2(1 - \delta_1) > \frac{1}{2}$: player 2 patient ($\delta_2 > \frac{1}{2}$) and player 1 impatient ($\delta_1 < 1 - \frac{1}{2\delta_2}$), because player 1's last-period power is worth little to an impatient player 1, while player 2's period-2 power is worth a lot to a patient player 2.

24 Muñoz-Garcia Ch. 6 Exercise 8

Advanced · block L4

Source: Muñoz-Garcia Ch. 6 Exercise 8, page 303 (statement in the book and in Exercise 24 of the list; not reproduced here).

What is asked. In three-player Rubinstein bargaining with unanimity (proposers rotate $1, 2, 3, 1, \ldots$; common $\delta \in (\frac{1}{2}, 1)$), show that stationary strategies give a unique SPE outcome and find each player's share.

Method. The stationarity argument of Rubinstein's two-player game (Tadelis §11.3, eqs. (11.4)–(11.5), pp. 228–229; Gibbons pp. 70–71): in a stationary equilibrium each proposer offers each responder exactly the discounted value of what the responder gets if the game moves on, and the values satisfy a linear system with a unique solution. Responders' votes are taken as pivotal (see the remark).

Solution. Notation. Players $j = 1, 2, 3$; $j + 1$ is the next proposer after j (mod 3). For a stationary profile, $V_i(j)$ is player i's payoff, valued at the start of a period in which j proposes. Write $a =$ the proposer's share, $b =$ the share of the next proposer, $d =$ the share of the player who proposes two periods later.

Step 1: responders. If j's proposal is rejected, play moves to a period in which $j + 1$ proposes, so responder i gets $\delta V_i(j + 1)$. A responder whose vote decides the outcome accepts x_i iff

$$x_i \;\geq\; \delta\, V_i(j+1). \tag{15}$$

Step 2: proposers agree at once. Offering each responder exactly $\delta V_i(j + 1)$ gives proposer j

$$1 - \delta \sum_{i \neq j} V_i(j+1) \;\geq\; 1 - \delta\,(1 - V_j(j+1)) \;>\; \delta\, V_j(j+1),$$

where the first step uses $\sum_i V_i(j + 1) \leq 1$ and the second uses $\delta < 1$. The right side is what j gets by having the proposal rejected, so every proposer makes the cheapest acceptable offer and it is accepted. With agreement in every period, $\sum_i V_i(j) = 1$ for every j and

$$V_i(j) = \delta\, V_i(j+1) \;\; (i \neq j), \qquad V_j(j) = 1 - \sum_{i \neq j} V_i(j). \tag{16}$$

Step 3: solve. Stationarity makes the system symmetric under the rotation, so $V_j(j) = a$, $V_{j+1}(j) = b$, $V_{j+2}(j) = d$ for every j. By (16), the next proposer $j + 1$ gets, as responder to j, δ times his own proposer share:

$$b = V_{j+1}(j) = \delta\, V_{j+1}(j+1) = \delta\, a,$$

and the player $j + 2$, who will be the next proposer's "next proposer", gets

$$d = V_{j+2}(j) = \delta\, V_{j+2}(j+1) = \delta\,(b) = \delta\,(\delta a) = \delta^2 a.$$

Substituting into $a + b + d = 1$:

$$a + (\delta a) + (\delta^2 a) = 1 \;\Rightarrow\; a = \frac{1}{1+\delta+\delta^2}, \quad b = \frac{\delta}{1+\delta+\delta^2}, \quad d = \frac{\delta^2}{1+\delta+\delta^2}.$$

The nine-equation system (16) is linear and has exactly this solution (no symmetry needs to be assumed; `check_B_L04.py` solves it without it). Hence the outcome is unique: in round $t = 0$ player 1 proposes

$$(x_1, x_2, x_3) = \Big(\frac{1}{1+\delta+\delta^2},\; \frac{\delta}{1+\delta+\delta^2},\; \frac{\delta^2}{1+\delta+\delta^2}\Big),$$

and players 2 and 3 accept. For example, $\delta = 0.8$: $(\frac{25}{61}, \frac{20}{61}, \frac{16}{61}) \approx (0.410, 0.328, 0.262)$. As $\delta \to 1$ each share tends to $\frac{1}{3}$.

Remark: the pivotal-vote assumption. With simultaneous votes and unanimity, the stationary profile "every responder rejects every offer" is also an SPE: given that the other responder rejects, a vote changes nothing, so rejecting is a best reply, and no agreement is ever reached (payoffs 0). Uniqueness therefore requires that responders vote as if pivotal (no weakly dominated votes), or that responses are sequential. Under either, (15) holds and the outcome above is the only one. The restriction $\delta > \frac{1}{2}$ plays no role in the stationary derivation, which holds for every $\delta \in (0, 1)$; the exercise imposes it because, without stationarity, a patient enough group can support many other divisions as SPE outcomes, which is why the stationarity restriction matters.

Interpretation. The proposer's advantage now has two layers: the next proposer can threaten to wait one period (δ), the third player only two (δ^2). Shares fall geometrically with the distance to one's own turn to propose. With two players the same logic gives $\frac{1}{1+\delta}$ and $\frac{\delta}{1+\delta}$.

Verification. `check_B_L04.py` solves the nine linear equations (16) with sympy: one solution, equal to $(1, \delta, \delta^2)/(1 + \delta + \delta^2)$ for every proposer, with the proposer's gain from agreement $(1 - \delta^3)/(1 + \delta + \delta^2) > 0$; it checks the $\delta = 0.8$ shares and the limit $\frac{1}{3}$, and at $\delta = 0.8$ it tries every proposal on a grid of step $\frac{1}{200}$ against the stationary acceptance rules: none beats $\frac{25}{61}$. *Comparison with Muñoz-Garcia's answer* (pp. 303–304): same shares and the same comparative statics. The book obtains $x_{\text{two}} \geq \delta^2 x_{\text{prop}}$ by comparing with the share that player gets "as a proposer two periods from now"; the binding comparison is with what he gets as next period's responder, $\delta\, x_{\text{next}}$, which equals $\delta^2 x_{\text{prop}}$ in a stationary equilibrium, so the result is unaffected. The book's answer does not address the "all reject" equilibrium of simultaneous voting described in the remark.

Answer. Under stationary strategies (with pivotal votes) the SPE outcome is unique: player 1 offers $(\frac{1}{1+\delta+\delta^2}, \frac{\delta}{1+\delta+\delta^2}, \frac{\delta^2}{1+\delta+\delta^2})$ in round 0 and players 2 and 3 accept at once. Each proposer keeps $\frac{1}{1+\delta+\delta^2}$, gives the next proposer δ times that and the third player δ^2 times that.